PSYCON: A Multimodal Wearable System for Continuous Psychological Condition Monitoring

PSYCON Project Team

Independent Research Prototype — Engineering Design Document

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Abstract

Background: Psychological well-being is influenced by physiological, behavioral, and environmental factors that are difficult to capture with single-modality tools such as questionnaires or isolated biosensors. **Objective:** This paper presents PSYCON, a two-module wearable research prototype designed to continuously acquire synchronized physiological, environmental, and speech-derived signals to support early identification of changes associated with psychological well-being. **Methods:** PSYCON consists of a Wrist Module for physiological sensing (galvanic skin response, photoplethysmography, motion, and temperature) and an Audio Module for speech and ambient-light sensing, each built around an ESP32 microcontroller with independent LiPo-battery power and USB-C charging. **Results:** We report the current, hardware-confirmed system architecture, sensor allocation, communication interfaces, and power subsystem design; items pending experimental validation — including the GSR analog front end, firmware integration, and electrical characterization — are explicitly identified. **Conclusions:** The modular, low-cost architecture of PSYCON provides a scalable foundation for multimodal mental-health data collection and future AI-driven analysis, while maintaining a clear distinction between confirmed engineering facts and items awaiting validation.

Keywords: wearable sensing; multimodal monitoring; psychological well-being; galvanic skin response; photoplethysmography; ESP32; mental health technology

Part I
Volume 1: Project Foundation

1 Project Overview

1.1 Project Identification

The system described in this document is named **PSYCON** (Psychological Condition Monitoring System). PSYCON is defined as a multimodal wearable system designed to continuously monitor physiological, environmental, and speech-derived indicators in order to assist in the early identification of changes associated with psychological well-being.

1.2 Executive Summary

PSYCON is a two-module wearable research prototype that combines physiological sensing, environmental monitoring, and speech analysis into a unified platform for continuous mental health assessment. Rather than relying on a single biomarker or questionnaire, the system captures multiple complementary signals—including cardiovascular activity, electrodermal activity, body temperature, motion, ambient light, and speech—to provide a richer understanding of a user's psychological state. The platform is intended as an assistive screening and monitoring tool for research and educational purposes, and it is designed to support long-term data collection, personalized analysis, and future integration with AI-driven mental health technologies.

1.3 Project Objective

The objective of the project is to develop a wearable system capable of continuously collecting multimodal data that may help identify patterns associated with psychological conditions, while remaining portable, modular, low-cost, and suitable for research and science-competition demonstrations.

1.4 Primary Goals

- Develop a wearable prototype using commercially available components.
- Acquire synchronized physiological, environmental, and audio data.
- Support continuous monitoring over extended periods.
- Enable future AI-based analysis of multimodal signals.
- Maintain a modular architecture that allows independent development and testing of subsystems.
- Keep the design scalable for future PCB integration.

1.5 Intended Applications

- Research in multimodal mental health monitoring.
- Longitudinal wellness tracking.
- Educational demonstrations.
- AI dataset collection.
- Human–computer interaction research.
- Science fair and engineering competitions.

1.6 High-Level Architecture

PSYCON consists of two independent wearable modules: a *Wrist Module*, responsible for physiological data acquisition, and an *Audio Module*, responsible for speech acquisition and environmental sensing. Figure 1 presents the overall system architecture, including sensor allocation, controller assignment, power subsystems, and the wireless data path to the host/analysis layer.

1.6.1 Wrist Module

Purpose. Physiological data acquisition.
Sensors.

- GSR electrodes (via ADS1115)
- MAX30102 pulse oximeter / heart-rate sensor
- MPU6050 inertial measurement unit
- MCP9808 digital temperature sensor

Controller. ESP32 DevKit (30-pin).
Power. Two 500 mAh LiPo batteries connected in parallel (≈1000 mAh total capacity), charged and protected via a TP4056 USB-C charging module.

1.6.2 Audio Module

Purpose. Speech acquisition and environmental sensing.
Sensors.

- INMP441 I^2S MEMS microphone
- TEMT6000 ambient light sensor

Controller. ESP32 DevKit (30-pin).
Power. One 500 mAh LiPo battery, charged and protected via a TP4056 USB-C charging module.

1.7 Data Collected

Table 1 summarizes the categories of data acquired by PSYCON across its two modules.

Table 1: Data modalities collected by PSYCON.

Category	Signals / Data
Physiological	Heart rate; pulse waveform (PPG); skin conductance (EDA/GSR); skin temperature; motion and orientation
Environmental	Ambient light level
Audio	Speech recordings for future AI feature extraction

1.8 Design Philosophy

The system follows five core engineering principles:

- **Modularity** — each subsystem can operate and be tested independently.
- **Scalability** — prototype hardware should transition cleanly to a custom PCB.
- **Maintainability** — use widely available, well-documented modules.
- **Low Cost** — balance performance with affordability.

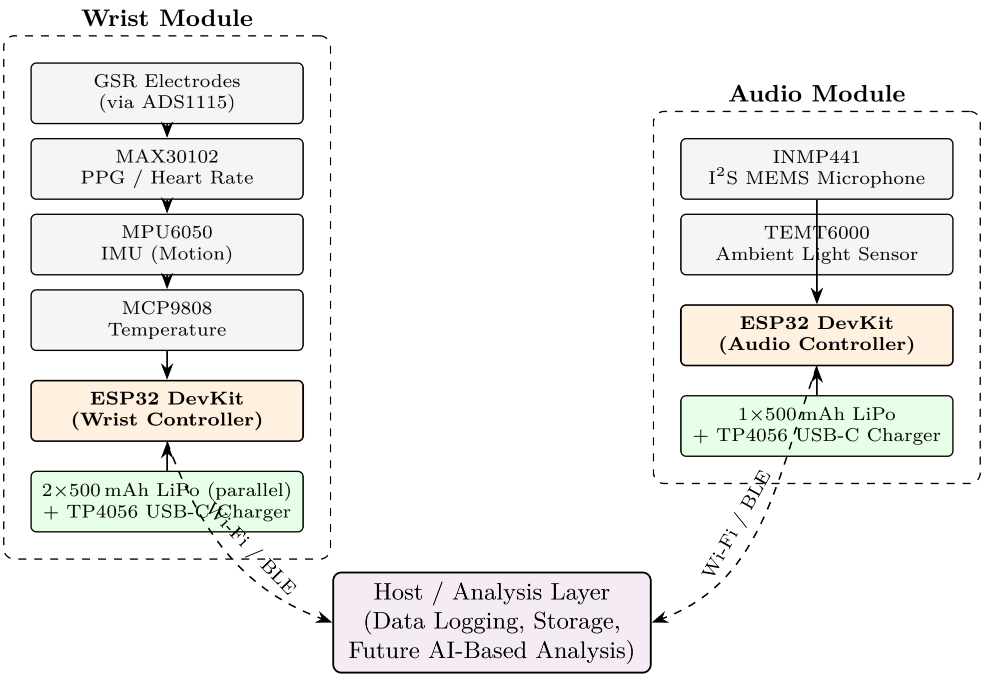

Figure 1: High-level system architecture of PSYCON, showing the Wrist Module (physiological sensing) and Audio Module (speech and environmental sensing) as independently powered ESP32-based subsystems that communicate wirelessly with a host/analysis layer.

- **Research Focus** — prioritize data quality and documentation over product miniaturization.

1.9 Current Hardware Configuration

Table 2 summarizes the current bill-of-materials-level hardware configuration for the PSYCON prototype, organized by controllers, power components, and sensors.

Table 2: Current hardware configuration of the PSYCON prototype.

Category	Component	Qty.
Controllers	ESP32 DevKit (30-pin)	2
Power	TP4056 USB-C charger module (protected)	2
	500 mAh 3.7 V LiPo battery (HLCY 651735P)	3
Sensors	MAX30102 pulse oximeter / heart-rate sensor	1
	ADS1115 16-bit ADC	1
	MPU6050 IMU	1
	MCP9808 temperature sensor	1
	TEMT6000 ambient light sensor	1
	INMP441 I^2S MEMS microphone	1
	GSR electrodes (analog front end TBD)	1 set

1.10 Communication Interfaces

Table 3 lists the communication interfaces used to connect each sensor to its respective ESP32 controller.

Table 3: Communication interfaces by sensor.

Interface	Connected Devices
I^2C	MAX30102, MPU6050, ADS1115, MCP9808
I^2S	INMP441
Analog	TEMT6000; GSR electrodes (through ADS1115)
Wi-Fi / BLE	ESP32 wireless communication (both modules)

1.11 Engineering Status

Validation

Confirmed:

- Component selection
- Module compatibility
- I^2C address compatibility
- Overall system architecture
- Power subsystem concept
- Sensor allocation between modules

Requirement

Pending validation:

- Final GSR analog front end
- Full firmware integration
- Electrical characterization (current, runtime, thermal)
- Long-duration testing

1.12 Assumptions and Scope

This document distinguishes between three categories of technical claims:

- **Confirmed information** — derived from hardware, datasheets, and documented design decisions.
- **Engineering assumptions** — based on standard module implementations.
- **Experimental items** — items that require physical testing before they can be claimed as verified.

Assumption

The distinction above is maintained throughout this document to preserve technical accuracy suitable for engineering documentation and science-competition review. Where a claim has not yet been experimentally validated, it is explicitly flagged rather than presented as a confirmed result.

2 System Requirements Specification

2.1 Purpose

This section defines exactly what PSYCON is expected to do from an engineering perspective. The requirements stated here act as the baseline against which the prototype will be evaluated.

2.2 Design Objectives

The system shall:

- Continuously collect physiological data.
- Continuously collect speech samples.
- Continuously monitor environmental conditions.
- Synchronize all collected data.
- Operate as a wearable prototype.
- Be modular.
- Be expandable.
- Be low-cost.
- Be reproducible using commercially available components.
- Support future AI-based analysis.

2.3 Functional Requirements

2.3.1 FR-1: Physiological Monitoring

The wrist module shall measure heart rate, pulse waveform (PPG), skin conductance (EDA/GSR), skin temperature, wrist motion, and wrist orientation.
Status: CONFIRMED Hardware selected PENDING Integration pending

2.3.2 FR-2: Audio Monitoring

The audio module shall capture speech, digitize audio using I^2S, stream or store recordings, and maintain synchronized timestamps.
Status: CONFIRMED Hardware selected PENDING Firmware pending

2.3.3 FR-3: Environmental Monitoring

The system shall measure ambient light for purposes of context awareness, sleep/environment correlation, and future feature extraction.
Status: CONFIRMED Hardware selected

2.3.4 FR-4: Data Synchronization

The system shall timestamp every sample, preserve acquisition order, and allow synchronization between the wrist and audio modules. This requirement is critical because machine-learning analysis becomes unreliable if physiological and audio events are misaligned in time.
Status: PENDING Firmware requirement

2.3.5 FR-5: Wireless Communication

Each ESP32 shall support Wi-Fi and BLE (future expansion), enabling data upload, remote monitoring, over-the-air (OTA) firmware updates (future), and local communication.
Status: CONFIRMED Supported by hardware

2.3.6 FR-6: Local Processing

Each module shall read sensors, filter signals, package data, detect sensor failures, and continue operation independently of the other module.

2.3.7 FR-7: Modular Operation

If one module fails, the second module shall continue operating. For example, an audio-module failure shall not stop physiological acquisition, and a physiological-module failure shall not stop audio acquisition.

2.3.8 FR-8: Expandability

Future sensors should be addable without redesigning the architecture. Candidate future sensors include ECG, EEG, EMG, respiration, skin hydration, and air-quality sensing.

2.4 Non-Functional Requirements

Reliability. The target is continuous operation without crashes. The prototype goal is a minimum continuous runtime of 6 h, with a stretch goal of 24 h.

Accuracy. Sensors should operate within manufacturer specifications; no artificial software correction should replace proper calibration.

Repeatability. Repeated measurements under identical conditions should produce similar results, which is especially important for GSR, heart rate, and temperature.

Maintainability. The system should use standard breakout boards, use labeled connectors, allow module replacement, and support firmware updates.

Portability. The system should fit on the body, operate on batteries, and avoid wired external power during use.

Scalability. Future revisions should migrate easily to a custom PCB, smaller ESP32 variants, and improved analog front ends.

Cost. The primary objective is high capability with affordable hardware; the approximate prototype cost target is lower than research-grade commercial systems.

Safety. The wearable shall use protected LiPo batteries, avoid exposed battery terminals, maintain a safe GSR excitation current, avoid charging while connected to GSR electrodes, and use insulated wiring.

2.5 Performance Targets

Table 4 summarizes the target operating behavior for each monitored parameter.

Table 4: Performance targets by parameter.

Parameter	Target
Heart-rate monitoring	Continuous
GSR monitoring	Continuous
Temperature monitoring	Continuous
Motion monitoring	Continuous
Ambient light	Continuous
Speech recording	Continuous or scheduled
Wireless communication	Stable
Battery-powered	Yes
Rechargeable	Yes

2.6 Hardware Constraints

Table 5 lists the hardware constraints of the finalized design, organized by controllers, power, and sensor allocation between the wrist and audio modules.

Table 5: Hardware constraints of the finalized PSYCON design.

Category	Components
Controllers	2 × ESP32 DevKit (30-pin)
Power	2 × TP4056 USB-C charger; 3 × 500 mAh LiPo battery
Wrist sensors	MAX30102; ADS1115; MCP9808; MPU6050; GSR electrodes
Audio sensors	INMP441; TEMT6000

2.7 Software Constraints

Firmware should:

- Support FreeRTOS tasks (recommended).
- Prevent blocking sensor reads.
- Handle communication failures gracefully.
- Log errors.
- Support firmware expansion.

2.8 Data Requirements

Every record should include a timestamp, module ID, sensor ID, sensor values, battery level (future), and error flags (future). Figure 2 illustrates the logical structure of a single data record as it is composed prior to storage or transmission.

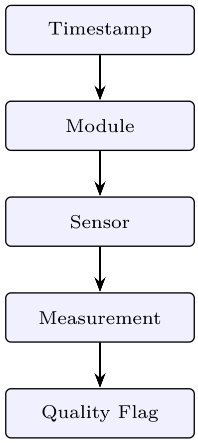

Figure 2: Logical structure of a single PSYCON data record, composed sequentially from acquisition timestamp through to a quality flag.

2.9 Assumptions

Assumption

The following are treated as current engineering assumptions until experimentally verified:

- Standard module revisions.
- Standard operating voltages.
- No I^{2}C address conflicts.
- Proper power distribution.
- Common ground maintained.
- Standard ESP32 DevKit V1 behavior.

2.10 Risks

Table 6 summarizes known engineering risks by priority level.

Table 6: Known engineering risks by priority.

Priority	Risk
HIGH	GSR analog front-end quality
HIGH	Battery runtime
HIGH	Charging while operating
HIGH	Long-term stability
MEDIUM	Audio noise
MEDIUM	Motion artifacts
MEDIUM	Synchronization drift
LOW	I^{2}C conflicts (none expected)
LOW	GPIO availability
LOW	Sensor compatibility

2.11 Acceptance Criteria

The prototype is considered functionally complete when:

- All sensors initialize successfully.
- No I^{2}C address conflicts occur.
- Audio records correctly.

- Physiological data streams continuously.
- Modules operate independently.
- Battery operation is stable.
- Data timestamps remain synchronized.
- Documentation is complete.
- Safety procedures are documented.
- Hardware passes integration testing.

2.12 Chapter Status

Validation

Completed for this chapter:

- Functional requirements
- Non-functional requirements
- Design objectives
- Constraints (hardware and software)
- Acceptance criteria
- Risk overview
- Data requirements

3 Overall System Architecture

3.1 System Overview

PSYCON is a distributed wearable sensing platform consisting of two independent electronic modules that work together to collect synchronized multimodal data for psychological state analysis. The architecture intentionally separates physiological sensing from speech sensing to reduce electrical noise, improve wearability, simplify mechanical design, and allow each subsystem to operate independently.

3.2 High-Level Architecture

Figure 3 presents the top-level data pipeline of PSYCON, from the two physical modules through synchronization, AI processing, and final psychological-condition assessment.

3.3 Module Responsibilities

3.3.1 Wrist Module

Purpose. Collect physiological signals directly from the user's body.

Primary functions.

- Heart-rate monitoring
- Pulse waveform acquisition
- Electrodermal activity (GSR)
- Skin temperature
- Motion sensing
- Timestamp generation
- Wireless communication

3.3.2 Audio Module

Purpose. Collect speech and environmental context.

Primary functions.

- Speech recording
- Ambient light monitoring
- Timestamp generation
- Wireless communication

3.4 Wrist Module Hardware

Controller. ESP32 DevKit (30-pin). Responsibilities include reading all sensors, managing I^2C communication, processing sensor data, buffering measurements, transmitting data, and managing power state.

MAX30102. Measures heart rate and pulse waveform (PPG) over an I^2C interface.

ADS1115. Provides high-resolution analog-to-digital conversion and is connected to the GSR analog front end.

Design Note

The ESP32's internal ADC has limited precision and non-linearity. The ADS1115 provides a 16-bit ADC with programmable gain, making it more suitable for low-level analog measurements such as GSR.

GSR electrodes. Measure changes in skin conductance associated with sweat-gland activity.

Requirement

Current status: electrodes selected; the analog front end is to be finalized and validated before human testing.

MPU6050. Measures acceleration and angular velocity, used for motion detection, activity recognition, and motion-artifact identification.

MCP9808. Measures skin temperature.

Design Note

The MCP9808 was selected because it provides higher accuracy and stability than common low-cost temperature sensors.

3.5 Audio Module Hardware

Controller. ESP32 DevKit (30-pin). Responsibilities include reading the microphone, reading the ambient light sensor, timestamping audio, and managing wireless communication.

INMP441. Measures digital audio over an I^2S interface and is used for speech feature extraction, voice-quality analysis, and future AI models.

TEMT6000. Measures ambient light intensity and is used for environmental context, sleep and activity correlation, and context-aware analysis.

3.6 Power Architecture

Each module operates independently, with its own battery, charging/protection circuit, and regulated rail. Figure 4 shows the power chain for both modules side by side.

The Wrist Module's parallel two-cell configuration yields an estimated battery capacity of approximately 1000 mAh, while the Audio Module operates from a single 500 mAh cell.

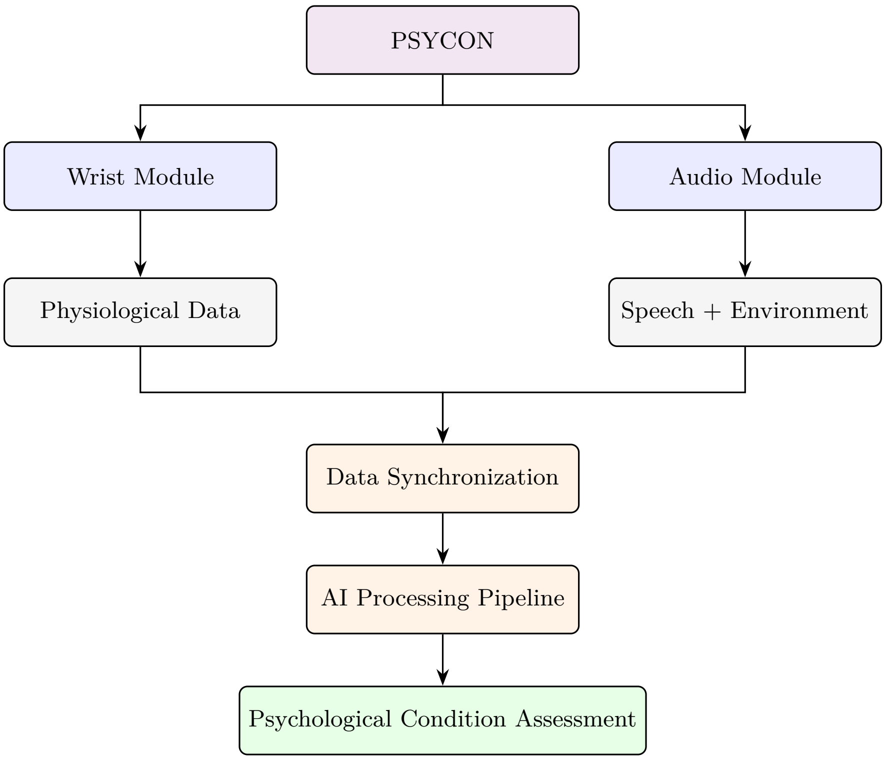

Figure 3: Top-level PSYCON data pipeline: independent acquisition by the Wrist and Audio Modules, followed by data synchronization, AI processing, and psychological-condition assessment.

3.7 Communication Architecture

3.7.1 I²C Bus (Wrist Module)

The I^{2}C bus on the Wrist Module is shared by the MAX30102, ADS1115, MCP9808, and MPU6050. Table 7 lists the expected device addresses; no address conflicts are expected.

Table 7: Expected I^{2}C device addresses on the Wrist Module bus.

Device	**Address**
MCP9808	`0x18`
ADS1115	`0x48`
MAX30102	`0x57`
MPU6050	`0x68` (default)

3.7.2 I²S Bus (Audio Module)

The I^{2}S bus connects the INMP441 microphone and carries the bit clock (SCK/BCLK), word select (WS/LRCLK), and serial data (SD) signals.

3.7.3 Analog Signals

Analog signals are handled through the ADS1115, including the GSR analog output. The TEMT6000 provides an analog output that may be connected either to an ESP32 ADC pin or to an ADS1115 channel if higher measurement precision is desired.

3.8 Data Flow

Figure 5 shows the end-to-end data flow from sensor acquisition through to AI processing. Each sample should receive its timestamp as close to acquisition as possible to minimize synchronization errors.

3.9 Power Flow

Figure 6 shows the generic power flow within each module, from battery through to the sensor loads. All sensors share a common ground within their respective module.

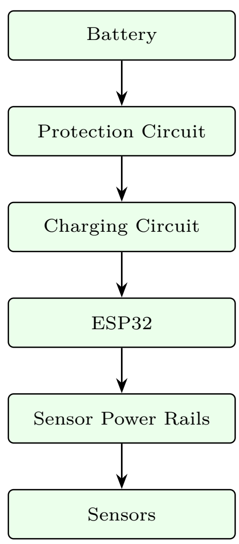

Figure 6: Generic per-module power flow, from battery through protection and charging circuitry to the microcontroller and sensor power rails.

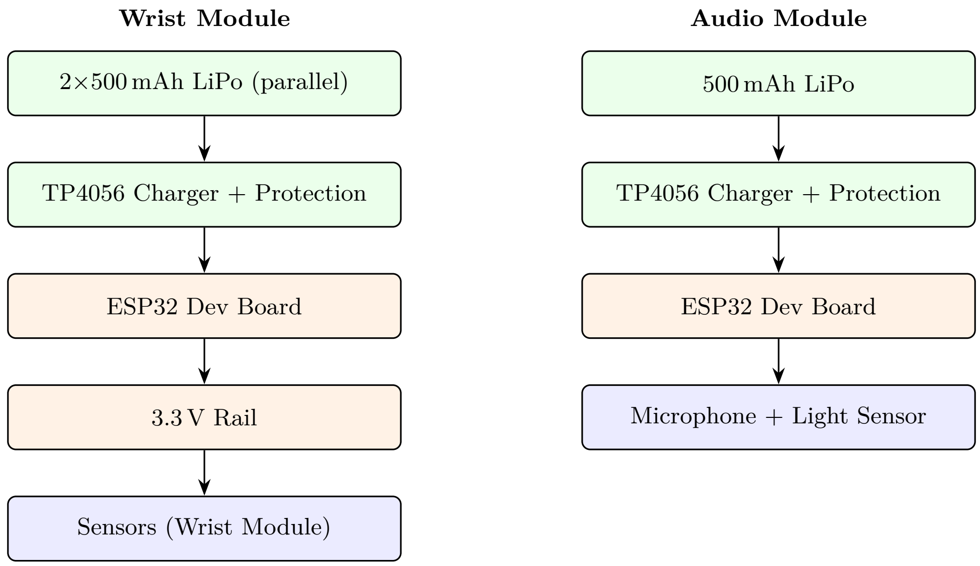

Figure 4: Independent power architecture for the Wrist Module (estimated battery capacity ≈1000 mAh) and the Audio Module, each with dedicated LiPo battery, TP4056 charge/protection circuit, and ESP32 controller.

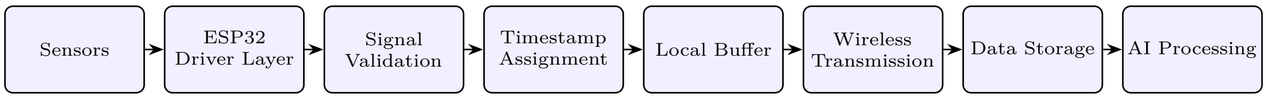

Figure 5: End-to-end PSYCON data flow, from raw sensor acquisition through driver-level reads, validation, timestamping, buffering, wireless transmission, storage, and AI processing.

3.10 Synchronization Strategy

Although physically separate, both modules should:

- Use a consistent time base.
- Include timestamps with every data packet.
- Include a module identifier.
- Include sensor identifiers.
- Handle temporary communication interruptions without losing acquisition order.

3.11 Fault Isolation

A failure in one module should not prevent the other from operating. For example, if the microphone fails, physiological sensing continues; if the wrist module resets, audio acquisition continues; and if wireless communication is unavailable, data should be buffered, subject to firmware support.

3.12 Engineering Advantages of the Split Architecture

Observation

- Reduced electrical interference between analog sensors and the microphone.
- Better weight distribution for wearability.
- Simpler wiring within each module.
- Independent testing and debugging.
- Easier future upgrades or PCB redesigns.
- Greater fault tolerance.

3.13 Current Limitations

Requirement

The following items remain to be validated experimentally:

- Final GSR analog front-end implementation.
- Battery runtime under real workloads.
- Continuous-operation stability.
- Thermal behavior during charging and prolonged operation.
- Wireless throughput and latency under sustained use.
- Long-term synchronization accuracy.

3.14 Chapter Status

Validation

Completed for this chapter:

- Overall system architecture
- Module separation rationale
- Hardware block architecture
- Data flow
- Power flow
- Communication architecture
- Synchronization strategy
- Fault isolation
- Engineering design rationale
- Current limitations

4 Complete Hardware Design

4.1 Hardware Design Philosophy

The hardware platform was designed with the following priorities:

- Research-grade sensor selection within a student budget.
- Modular architecture for independent testing.
- Commercially available components.
- Easy replacement and maintenance.
- Future migration to a custom PCB.
- Low-voltage battery-powered operation.
- Standard communication protocols (I^2C, I^2S, ADC).

4.2 Complete Bill of Hardware

Table 8 lists the complete hardware inventory for the PSYCON prototype, including quantity, module assignment, and function of each component.

4.3 ESP32 DevKit (30-pin)

Role. Primary microcontroller responsible for sensor acquisition, data processing, wireless communication, timestamp generation, power management, and firmware execution.

Table 9 lists the key specifications of the ESP32 DevKit.

Table 9: Key specifications of the ESP32 DevKit (30-pin).

Parameter	Value
MCU	ESP32-WROOM-32
CPU	Dual-core Xtensa LX6
Clock	Up to 240 MHz
SRAM	520 KB
Flash	Typically 4 MB
Logic voltage	3.3 V
Wi-Fi	802.11 b/g/n
Bluetooth	Classic + BLE
ADC	12-bit (used sparingly)
DAC	2 channels
GPIO	Up to 34 usable

Why it was selected. Integrated Wi-Fi and BLE; large community support; mature software ecosystem; multiple communication interfaces; sufficient processing capability; excellent cost-to-performance ratio.

Interfaces used.

- I^2C — MAX30102, MPU6050, ADS1115, MCP9808
- I^2S — INMP441
- Analog — optional TEMT6000 input; battery monitoring (future)

Engineering Note

The ESP32 operates at 3.3 V logic and its GPIOs are not 5 V tolerant. The internal ADC is adequate for basic measurements, but the ADS1115 is preferred for precision analog sensing.

4.4 MAX30102

Function. Photoplethysmography (PPG) sensor for cardiovascular monitoring.

Measurements. Heart rate; pulse waveform; SpO_2 capability (not currently a primary feature).

Interface. I^2C. **Default address:** `0x57`.

Table 10 lists the electrical characteristics of the MAX30102.

Table 10: Electrical characteristics of the MAX30102.

Parameter	Value
Supply	3.3–5 V (module)
Logic	3.3 V
LEDs	Red + IR
Internal ADC	Yes

Selection rationale. Widely used and documented; integrated optics and analog front end; compact and wearable-friendly.

Integration notes. Must sit flush against the skin; minimize ambient light leakage; reduce motion artifacts through mechanical design.

4.5 ADS1115

Function. High-resolution external ADC. **Purpose in PSYCON:** primary interface for GSR measurement.

Table 11 lists the key specifications of the ADS1115.

Table 11: Key specifications of the ADS1115.

Parameter	Value
Resolution	16-bit
Channels	4
Interface	I^2C
Supply	2.0–5.5 V
PGA	Yes

Default address. `0x48`, configurable up to `0x49`, `0x4A`, `0x4B`.

Why it was selected. Better linearity and effective resolution than the ESP32 ADC; programmable gain supports low-level analog signals; well supported by existing libraries.

Design Note

The ADS1115 is intended for the GSR analog front end. Its additional channels remain available for future analog sensors.

4.6 GSR Subsystem

Purpose. Measure electrodermal activity (skin conductance).

Current hardware. ADS1115; electrodes.

Requirement

Current status: prototype-level implementation.

Table 8: Complete bill of hardware for the PSYCON prototype.

Component	Qty.	Module	Function
ESP32 DevKit (30-pin)	2	Both	Main controller
TP4056 USB-C charger (HW-373 V1.2.1, protected)	2	Both	Battery charging and protection
500 mAh 3.7 V LiPo (HLCY 651735P)	3	Both	Power source
MAX30102	1	Wrist	Heart rate & PPG
ADS1115	1	Wrist	16-bit ADC
GSR electrodes	1 pair	Wrist	Skin conductance
MPU6050	1	Wrist	Motion sensing
MCP9808	1	Wrist	Temperature
INMP441	1	Audio	Digital microphone
TEMT6000	1	Audio	Ambient light

Future Work

Planned improvements:

- Constant-current excitation source.
- Patient-protection resistor(s).
- RC low-pass filtering.
- Improved analog conditioning.
- Stable reference.
- Shielded wiring if necessary.

Engineering Note

The quality of GSR measurements depends primarily on the analog front end rather than the ADC alone. A documented and validated front end will be important for research-quality data.

4.7 MPU6050

Function. Six-axis inertial measurement unit.

Measurements. 3-axis acceleration; 3-axis angular velocity.

Interface. I^2C. **Address:** default `0x68`, alternate `0x69`.

Uses in PSYCON. Motion detection; activity recognition; identifying motion artifacts in physiological signals.

Why it was selected. Mature and inexpensive; widely supported; adequate performance for wearable motion sensing.

4.8 MCP9808

Function. Digital temperature sensor. **Measurement:** skin temperature.

Interface. I^2C. **Default address:** `0x18`, configurable through address pins.

Table 12 lists the key specifications of the MCP9808.

Table 12: Key specifications of the MCP9808.

Parameter	Value
Accuracy	±0.25 °C (typical)
Supply	2.7–5.5 V
Resolution	Up to 0.0625 °C

Selection rationale. Chosen for higher accuracy and stability compared with lower-cost digital temperature sensors.

4.9 INMP441

Function. Digital MEMS microphone. **Interface:** I^2S. **Measurements:** digital audio for speech analysis.

Table 13 lists the specifications of the INMP441.

Table 13: Specifications of the INMP441 digital microphone.

Parameter	Value
Supply	1.8–3.3 V
Output	24-bit I^2S
Directivity	Omnidirectional

Selection rationale. Native I^2S interface; low noise; good compatibility with the ESP32.

Integration notes. Keep away from switching power circuitry; ensure the acoustic port is unobstructed; verify gain, clipping margin, and DMA stability in firmware.

4.10 TEMT6000

Function. Ambient light sensor. **Output:** analog voltage proportional to light intensity. **Purpose:** provide environmental context (e.g., lighting conditions) for data interpretation.

Table 14 lists the electrical characteristics of the TEMT6000.

Table 14: Electrical characteristics of the TEMT6000.

Parameter	Value
Supply	3.3–5 V
Output	Analog

Design Note

For higher precision, the TEMT6000 output may be routed through the ADS1115 rather than the ESP32's internal ADC.

4.11 TP4056 USB-C Charger (HW-373 V1.2.1)

Function. Single-cell Li-ion/LiPo charging module with battery protection.

Confirmed features. USB-C input; battery protection; overcharge protection; over-discharge protection; over-current protection; short-circuit protection.

Expected components. TP4056 charger IC; DW01A protection IC; FS8205A dual MOSFET. IC markings should be verified from the physical board.

Warning

Limitations:

- No true power-path / load-sharing.
- Not recommended to operate the wearable while charging.
- Reverse-polarity protection is not guaranteed.

4.12 LiPo Battery

Model. HLCY 651735P. Table 15 lists its specifications.

Table 15: Specifications of the HLCY 651735P LiPo battery.

Parameter	Value
Chemistry	Lithium polymer
Nominal voltage	3.7 V
Fully charged	4.2 V
Capacity	500 mAh

Configuration. Wrist Module: 2 × 500 mAh in parallel, total nominal capacity ≈1000 mAh. Audio Module: 1 × 500 mAh.

Warning

Safety notes:

- Never connect the wrist batteries in series.
- Inspect for swelling, punctures, or damaged insulation before use.
- Disconnect charging before attaching GSR electrodes to a participant.

4.13 Hardware Compatibility Summary

Table 16 summarizes pairwise compatibility between the ESP32 controller and each connected sensor, together with bus-level conflict checks.

4.14 Remaining Hardware Validation

Requirement

The following still require experimental verification before final sign-off:

- ESP32 board revision and regulator identification.
- TP4056 PROG resistor value and actual charge current.
- Battery discharge characteristics under load.
- GSR analog front-end implementation and calibration.
- Current consumption in all operating modes.
- Thermal performance during charging and continuous operation.
- Brownout behavior and low-battery handling.
- Final wiring verification after assembly.

4.15 Chapter Status

Validation

Completed for this chapter:

- Full hardware inventory
- Detailed description of every module
- Electrical specifications
- Selection rationale
- Integration notes
- Power subsystem
- Safety considerations
- Compatibility review
- Remaining validation tasks

5 Electrical Design & Complete Wiring

5.1 Electrical Design Philosophy

The electrical system is designed around two independent ESP32-based modules: the **Wrist Module** (physiological sensing) and the **Audio Module** (speech sensing). Each module has an independent battery, independent charger, independent ESP32, and independent firmware. This separation minimizes cross-module interference, improves fault tolerance, and simplifies debugging.

5.2 Power Architecture

Figure 7 shows the complete power chain for both modules, from battery cells through regulation to the connected sensor loads.

Warning

Many ESP32 DevKit boards expect a regulated 5 V on `VIN` or USB, not a raw LiPo voltage. If powering directly from a LiPo cell, an appropriate regulator or a board designed for single-cell LiPo input is required. This must be verified on the specific board used.

5.3 Wrist Module GPIO Assignment

Table 17 lists the GPIO assignment for the Wrist Module.

Table 17: Wrist Module ESP32 GPIO assignment.

ESP32 GPIO	Device / Function
GPIO21	I^{2}C SDA — MAX30102, MPU6050, ADS1115, MCP9808
GPIO22	I^{2}C SCL — MAX30102, MPU6050, ADS1115, MCP9808
GPIO34	ADS1115 `ALERT` (optional interrupt)
GPIO35	Reserved
GPIO32	Future battery monitor
GPIO33	Future interrupt

Power. All Wrist Module sensors (MAX30102, ADS1115, MCP9808, MPU6050) are supplied from the

Table 16: Hardware compatibility summary.

Component Pair / Check	Status
ESP32 ↔ MAX30102	COMPATIBLE
ESP32 ↔ MPU6050	COMPATIBLE
ESP32 ↔ ADS1115	COMPATIBLE
ESP32 ↔ MCP9808	COMPATIBLE
ESP32 ↔ INMP441	COMPATIBLE
ESP32 ↔ TEMT6000	COMPATIBLE
I^2C address conflicts	NONE EXPECTED
Logic-level conflicts	NONE EXPECTED (3.3 V operation)

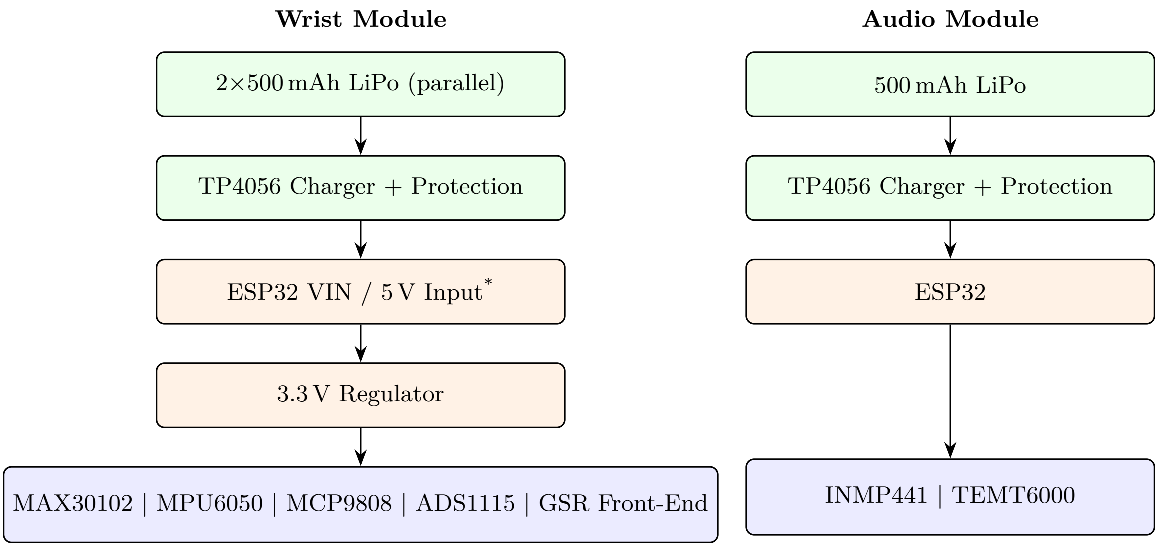

Figure 7: Complete power distribution for the Wrist Module and Audio Module, from LiPo cells through the TP4056 charger/protection stage, ESP32 power input, regulation, and final sensor loads. *Final power wiring depends on the exact ESP32 board; see the note below.

3.3 V rail.

Ground. A common ground is shared by every module.

5.4 Audio Module GPIO Assignment

Table 18 lists the GPIO assignment for the Audio Module.

Table 18: Audio Module ESP32 GPIO assignment.

ESP32 GPIO	Device / Function
GPIO26	I^2S BCLK
GPIO25	I^2S LRCLK (WS)
GPIO33	I^2S DATA
GPIO32	TEMT6000 analog output
3.3 V	Sensor VCC
GND	Common ground

5.5 I^2C Bus

Figure 8 shows the shared I^2C bus topology on the Wrist Module. Table 19 lists the expected device addresses; no address conflicts are expected.

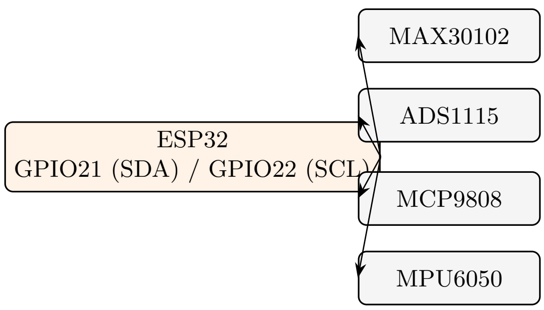

Figure 8: Shared I^2C bus topology on the Wrist Module: the ESP32 (GPIO21/SDA, GPIO22/SCL) drives MAX30102, ADS1115, MCP9808, and MPU6050 on a single bus.

Table 19: Expected I^2C device addresses on the Wrist Module bus.

Device	Address
MCP9808	`0x18`
ADS1115	`0x48`
MAX30102	`0x57`
MPU6050	`0x68`

5.6 I²S Bus

The Audio Module I²S bus connects the ESP32 to the INMP441 digital microphone as follows: GPIO26 → BCLK, GPIO25 → WS, GPIO33 → SD, 3.3 V → VDD, GND → GND. The INMP441 L/R select pin is connected to GND, selecting the default left channel.

5.7 ADS1115 Connections

Table 20 lists the complete ADS1115 wiring.

Table 20: ADS1115 pin connections.

ADS1115 Pin	Connection
VDD	3.3 V
GND	GND
SDA	GPIO21
SCL	GPIO22
ADDR	GND (address `0x48`)
ALRT	Optional, GPIO34
A0	GSR front end
A1–A3	Reserved

5.8 MAX30102 Connections

Table 21 lists the complete MAX30102 wiring.

Table 21: MAX30102 pin connections.

Sensor Pin	ESP32 Connection
VIN	3.3 V
GND	GND
SDA	GPIO21
SCL	GPIO22
INT	Optional, GPIO27

5.9 MPU6050 Connections

Table 22 lists the complete MPU6050 wiring. The device address is `0x68`.

Table 22: MPU6050 pin connections.

Pin	ESP32 Connection
VCC	3.3 V
GND	GND
SDA	GPIO21
SCL	GPIO22
INT	Optional, GPIO14
AD0	GND

5.10 MCP9808 Connections

Table 23 lists the complete MCP9808 wiring. The device address is `0x18`.

Table 23: MCP9808 pin connections.

Pin	ESP32 Connection
VIN	3.3 V
GND	GND
SDA	GPIO21
SCL	GPIO22
ALERT	Optional, GPIO13
A0–A2	GND

5.11 TEMT6000 Connections

Table 24 lists the complete TEMT6000 wiring.

Table 24: TEMT6000 pin connections.

Pin	ESP32 Connection
VCC	3.3 V
GND	GND
SIG	GPIO32 (ADC)

5.12 GSR Front End (Current Prototype)

Figure 9 shows the current prototype signal chain for electrodermal activity acquisition.

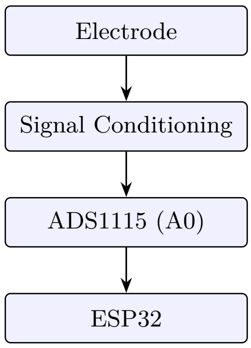

Figure 9: Current prototype GSR signal chain: electrode → signal conditioning → ADS1115 channel A0 → ESP32.

Requirement

Current status: prototype implementation.

Future Work

Planned improvements:

- Constant-current source.
- RC filter.
- Patient protection resistor.
- Shielding.
- Calibration procedure.

5.13 Grounding Strategy

Engineering Note

Each module shall use a single common ground. Ground loops should be avoided, analog return paths kept short, and the microphone return path kept away from high-current power traces where possible.

5.14 Decoupling

Design Note

A 0.1 µF ceramic capacitor is recommended close to each sensor supply, together with a 10 µF bulk capacitor near the ESP32 power input. Additional bulk capacitance should be added if Wi-Fi brownouts are observed. Some breakout boards already include these components.

5.15 Wiring Recommendations

- Use 26–28 AWG stranded wire for power.
- Use 28–30 AWG stranded wire for signal wiring.
- Keep I^2C runs short.
- Use twisted-pair or closely routed GSR leads if cable length increases.
- Provide secure strain relief for battery and sensor connections.

5.16 Electrical Safety

Warning

- Never apply 5 V directly to ESP32 GPIO pins.
- Verify battery polarity before connecting.
- Do not charge the module while connected to GSR electrodes.
- Insulate all exposed battery terminals.
- Disconnect power before rewiring.
- Inspect LiPo cells before every test.

5.17 Estimated Current Budget

Table 25 lists the estimated per-device current draw for both modules. The Wrist Module total is estimated at ≈90–210+ mA, and the Audio Module total at ≈85–205+ mA.

Table 25: Estimated current budget by module and device.

Module	Device	Approx. Current
Wrist	ESP32	80–200 mA (Wi-Fi/CPU dependent)
Wrist	MAX30102	1–5 mA
Wrist	MPU6050	3–5 mA
Wrist	ADS1115	<1 mA
Wrist	MCP9808	<1 mA
Wrist	GSR front end	Depends on final circuit
Audio	ESP32	80–200 mA
Audio	INMP441	1–2 mA
Audio	TEMT6000	<1 mA

5.18 Remaining Electrical Validation

Requirement

The following must still be verified on the assembled hardware:

- Exact ESP32 power input path and acceptable VIN range.
- 3.3 V rail stability under Wi-Fi load.
- Battery runtime under continuous operation.
- Charge current of the TP4056 (PROG resistor).
- Brownout threshold.
- Peak current during wireless transmission.
- Thermal performance during charging.
- Final GPIO validation with firmware.

5.19 Chapter Status

Validation

Completed for this chapter:

- Complete wiring architecture
- GPIO assignments
- I^2C wiring
- I^2S wiring
- Power distribution
- Grounding strategy
- Current budget
- Electrical safety
- Integration notes

6 Firmware Architecture & Software Design

6.1 Firmware Overview

PSYCON uses two independent ESP32 firmware projects, one per module.

Wrist Module firmware is responsible for: MAX30102 acquisition; MPU6050 acquisition; MCP9808 acquisition; GSR acquisition (via the ADS1115); timestamping; local preprocessing; BLE/Wi-Fi transmission; battery monitoring; and error logging.

Audio Module firmware is responsible for: INMP441 audio capture; TEMT6000 acquisition; audio buffering; voice preprocessing; BLE/Wi-Fi transmission; battery monitoring; and error logging.

6.2 Software Layers

Figure 10 shows the layered firmware architecture shared by both modules, from the application layer down to the physical ESP32 hardware.

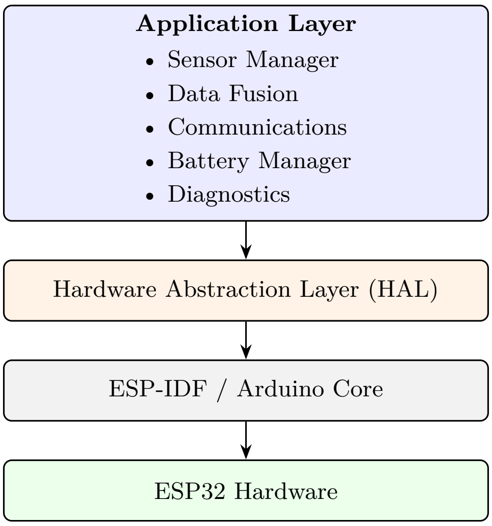

Figure 10: Layered firmware architecture: an application layer (sensor management, data fusion, communications, battery management, diagnostics) sits above a hardware abstraction layer, the ESP-IDF/Arduino core, and the ESP32 hardware itself.

6.3 Wrist Module Tasks

The Wrist Module firmware is organized into eight cooperating tasks, summarized in Table 26 and illustrated in Figure 11.

6.4 Audio Tasks

The Audio Module firmware runs four cooperating tasks: the Microphone Task, the Light Sensor Task, the Packet Builder, and the Battery Monitor / Communication task. Figure 12 shows the audio acquisition pipeline from initialization through transmission.

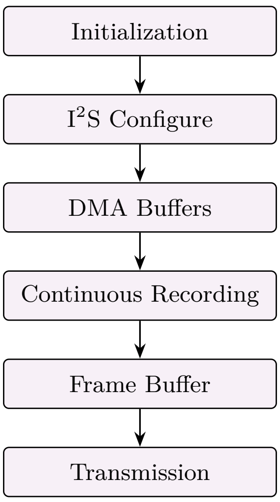

Figure 12: Audio Module acquisition pipeline: initialization, I^2S configuration, DMA buffer setup, continuous recording, frame buffering, and transmission.

6.5 Data Flow

Figure 13 shows the end-to-end firmware data flow, from raw sensor reads through to server-side AI processing.

6.6 Packet Structure

Table 27 lists an example physiological packet structure (Wrist Module), and Table 28 lists the audio packet structure (Audio Module).

Table 27: Example physiological (Wrist Module) packet fields.

#	Field
1	Timestamp
2	Heart rate
3	SpO_2
4	Accelerometer X
5	Accelerometer Y
6	Accelerometer Z
7	Gyroscope X
8	Gyroscope Y
9	Gyroscope Z
10	Temperature
11	GSR
12	Battery
13	Checksum

Table 28: Audio packet fields.

#	Field
1	Timestamp
2	PCM samples
3	Battery
4	Checksum

6.7 Sampling Strategy

Table 29 summarizes the acquisition regime for each signal. Actual sampling frequencies are to be finalized based on testing and downstream processing requirements.

Table 29: Sampling strategy by signal.

Signal	Regime
Heart	Continuous
Motion	Continuous
Temperature	Periodic
Light	Periodic
Audio	Continuous
GSR	Continuous

Requirement

Actual sampling frequencies for each signal remain to be finalized based on empirical testing and the requirements of downstream processing.

6.8 Error Handling

Algorithm 1 describes the generic sensor error-handling procedure used across both firmware projects. The system

Table 26: Wrist Module firmware task summary.

Task	Trigger	Responsibilities
1. Sensor Initialization	Runs once	Initializes MAX30102, ADS1115, MPU6050, MCP9808
2. Heart Rate Task	Periodic	Reads RED LED; reads IR LED; filters signal; stores buffer
3. Motion Task	Periodic	Reads accelerometer and gyroscope; computes movement and orientation
4. Temperature Task	Periodic	Reads MCP9808; stores temperature
5. GSR Task	Periodic	Reads ADS1115; converts ADC value to conductance; stores value
6. Packet Builder	Periodic	Assembles one packet from timestamp, heart, motion, temperature, GSR, and checksum
7. BLE/Wi-Fi Task	Periodic	Uploads packets; retries on failure
8. Battery Monitor	Periodic	Checks battery voltage; issues low-battery / shutdown warnings

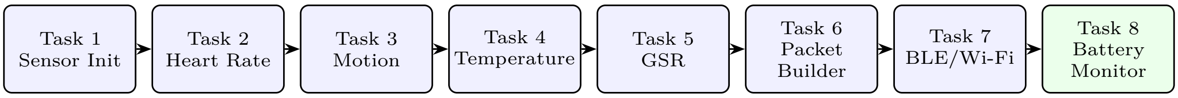

Figure 11: Wrist Module task pipeline: one-time sensor initialization followed by periodic acquisition tasks (heart rate, motion, temperature, GSR), packet assembly, wireless upload, and battery monitoring.

Algorithm 1: Generic Sensor Error Handling

```
Data: Sensor read request
Result: Sensor value or logged failure, without halting
        the system
Attempt sensor read;
if timeout or communication error detected then
    Retry read;
    if retry fails then
        Log error (sensor ID, error type, timestamp);
Continue acquisition for all remaining sensors;
```

is designed so that a single failed sensor never causes a complete crash.

6.9 Communication

Every transmitted packet contains: a timestamp; a sensor ID; the payload; a CRC/checksum; and a sequence number. Because each packet carries a sequence number, missing packets can be detected at the receiving end.

6.10 Logging

The firmware maintains logs for: sensor initialization failures; I^{2}C communication errors; buffer overruns; Wi-Fi/BLE reconnects; battery warnings; brownout events; and memory allocation failures.

6.11 Battery Management

Figure 14 shows the firmware battery-management state progression. Sudden power loss is avoided where possible by saving pending data before shutdown.

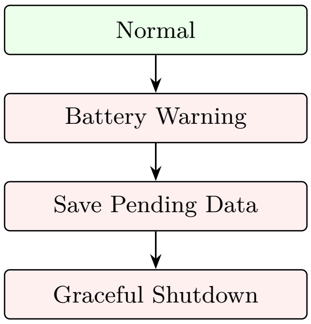

Figure 14: Firmware battery-management state progression, from normal operation through a battery-warning state, saving of pending data, and graceful shutdown.

6.12 Boot Sequence

Figure 15 shows the firmware boot sequence, from power-on to continuous acquisition.

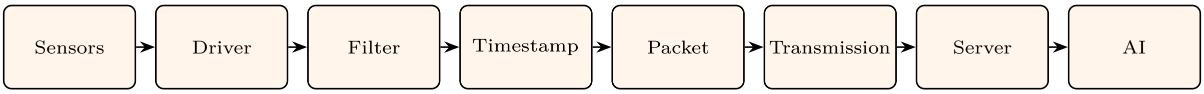

Figure 13: End-to-end firmware data flow, from sensor acquisition through driver-level reads, filtering, timestamping, packet assembly, wireless transmission, server ingestion, and AI processing.

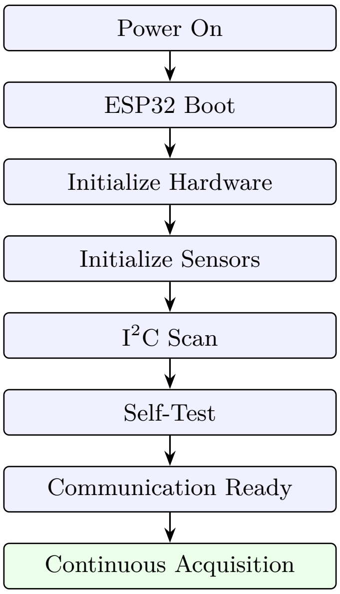

Figure 15: Firmware boot sequence, from power-on through hardware and sensor initialization, an I^2C bus scan, self-test, and communication readiness, to continuous acquisition.

6.13 Watchdog Recovery

Engineering Note

If the firmware hangs, the watchdog resets the ESP32. The boot reason is logged, and data acquisition resumes automatically after restart.

6.14 Memory Management

Design Note

Firmware uses fixed-size buffers and circular/ring buffers for continuous streams. Dynamic memory allocation (`malloc`/`new`) is avoided within continuous acquisition loops to reduce heap fragmentation.

6.15 Audio Buffering

Figure 16 shows the audio buffering chain, which helps prevent sample loss during temporary communication delays.

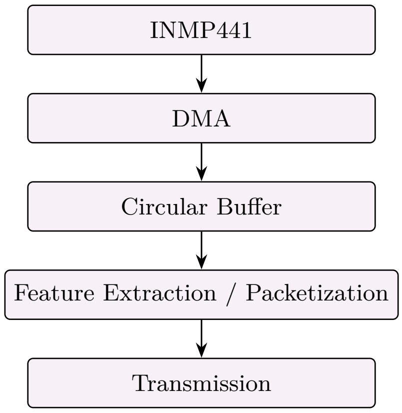

Figure 16: Audio buffering chain from the INMP441 microphone through DMA and a circular buffer to feature extraction/packetization and transmission, reducing sample loss during transient communication delays.

6.16 Firmware Folder Structure

Listing 1 shows the planned firmware repository layout, shared between the Wrist and Audio Module projects.

```
firmware/
|-- wrist/
|   |-- sensors/
|   |-- communication/
|   |-- battery/
|   |-- logging/
|   |-- utilities/
|   `-- main.cpp
|-- audio/
|   |-- microphone/
|   |-- communication/
|   |-- battery/
|   |-- logging/
|   `-- main.cpp
`-- shared/
    |-- config.h
    |-- packet.h
    |-- crc.cpp
    `-- timestamp.cpp
```

Listing 1: PSYCON firmware repository structure.

6.17 Firmware Validation Checklist

Requirement

Before full integration, the following must be verified:

- Sensor initialization succeeds.
- All I^2C devices are detected.
- No address conflicts.
- Stable sampling over extended periods.
- No packet corruption.
- No memory leaks.
- No watchdog resets during normal operation.

- Battery warnings trigger correctly.
- Graceful shutdown works.
- Audio stream remains continuous without excessive buffer overruns.

6.18 Version Control

The following are tracked under version control: firmware version; hardware revision; configuration changes; sensor calibration updates; release notes; and test results for each firmware version.

6.19 Chapter Status

Validation

Completed for this chapter:

- Firmware architecture
- Task scheduling
- Sensor acquisition flow
- Packet structure
- Communication design
- Logging strategy
- Battery management
- Error handling
- Memory management
- Boot process
- Validation checklist

7 AI Pipeline, Backend Architecture & Machine Learning Design

7.1 AI System Overview

PSYCON consists of two synchronized AI pipelines whose feature sets are fused into a final inference engine.

Physiological AI Pipeline. Heart rate; heart rate variability (HRV); blood-oxygen trends (SpO_2, if used); galvanic skin response (GSR/EDA); body temperature; motion/activity.

Speech AI Pipeline. Voice recording; speech characteristics; language features; conversation analysis; acoustic features.

7.2 Overall Architecture

Figure 17 shows how physiological features from the Wrist Module and speech features from the Audio Module converge in the AI fusion engine.

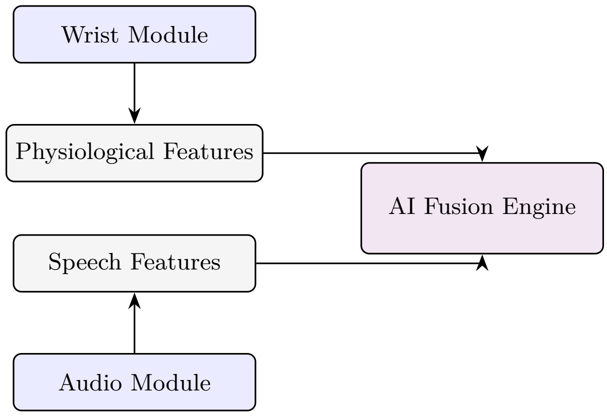

Figure 17: Overall AI architecture: physiological features from the Wrist Module and speech features from the Audio Module converge in the AI fusion engine.

7.3 Data Acquisition

Physiological. Heart rate, HRV, GSR, temperature, and motion are collected continuously. Every sample receives a timestamp, sensor ID, quality flag, and battery status.

Speech. The target system collects consented speech continuously in an approved study. The current firmware has not yet demonstrated continuous capture; Figure 18 shows the intended acquisition pipeline.

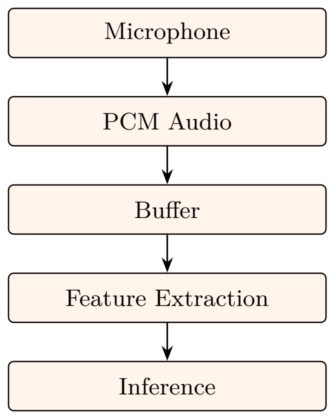

Figure 18: Speech acquisition pipeline: microphone → PCM audio → buffer → feature extraction → inference.

7.4 Feature Engineering

Table 30 summarizes candidate features under consideration for each signal modality.

Research Note

The feature lists above represent candidate features under consideration. Final feature selection should be informed by empirical validation on collected data.

7.5 Implemented Audio Engineering Baseline

The software implements one deterministic mono PCM16 extractor named `psycon_audio`. Its ordered feature vector contains duration, RMS and peak dBFS, clipping fraction, DC offset, zero-crossing rate, spectral centroid, 85% spectral rolloff, spectral flatness, mean/standard-deviation/minimum/maximum autocorrelation pitch from 70–400 Hz, relative pitch jitter, bounded voice stability,

Table 30: Candidate features by signal modality.

Modality	**Candidate Features**
Heart	Mean HR; HR variability; resting trend; peak frequency; recovery trend
Motion	Activity level; tremor; stability; walking; rest periods
Temperature	Mean; drift; sudden changes; long-term baseline
GSR	Baseline conductance; slow tonic changes; rapid phasic responses; peak count; recovery slope
Speech (acoustic)	Speaking rate; pause duration; pitch statistics; energy/loudness; voice stability; spectral features (e.g., MFCCs)
Speech (language)	Vocabulary diversity; sentence length; sentiment; emotion-related language; topic transitions

energy-active frame fraction, pause count, and pause duration. It has no downloaded-model or third-party feature-extractor dependency. Feature changes must update the contract and equivalence tests together.

Every extracted window retains the session and device identifiers, packet sequence, device timestamp, first sample index, sample count, nominal sample rate, SHA-256 source-packet digest, and extractor identity. Before inference, the pipeline returns one explicit state: usable, insufficient audio, clipped, too noisy, corrupt, or missing audio. Missing input is never replaced with zero-valued audio.

Deterministic fixtures cover silence, impulse, a 220 Hz tone, clipped tone, amplitude-modulated speech-like input, and seeded white noise. The local demo passes each fixture through the Protocol v2 decoder and writes a JSON decision report. These tests verify the software contract and abstention behavior only; final thresholds require evidence from the installed INMP441, and continuous capture, light alignment, clock behavior, and transport-loss recovery remain physical integration gates.

The consent-gated `psycon_transcription` stage transcribes contiguous acoustic windows marked usable with a locally cached faster-whisper model; rejected windows never silently enter speech-to-text processing. Results preserve timestamped transcript segments, source-sample boundaries, detected language, approximate confidence, engine/model identity, and explicit complete, no-speech, skipped-quality, unavailable, not-requested, or failed states. Recordings are not sent to a third-party transcription API.

The development profile uses the multilingual `small` model on CPU with int8 computation, replacing `tiny` because transcript errors directly contaminate downstream language features. The approved production profile runs multilingual `turbo` with float16 computation on a CUDA GPU inside the PSYCON backend. The model, device, and compute type are deployment configuration and must be recorded with every result.

Before release, compare `small`, `medium`, and `turbo` on representative consented multilingual recordings. Report per-language word or character error rate, language-detection accuracy, end-to-end latency, peak memory, failure rate, and stability of downstream language features. Freeze the selected model only after this evidence is recorded; a server deployment does not waive latency, concurrency, cost, or privacy constraints.

The `psycon_language` baseline implements the English features specified in this chapter: vocabulary diversity, sentence count and length, transparent lexicon-based sentiment, emotion-related word counts, topic-transition distance, speech duration, pause duration, and speaking rate. The transcription layer accepts multilingual speech, but the language-feature extractor explicitly abstains for non-English transcripts until validated language-specific tokenization and lexicons exist. Insufficient text and unavailable transcription also abstain explicitly. These features are intended for controlled research comparison; they are not validated emotion recognition or clinical interpretation. Filler-word classification is outside the Week 3 scope.

7.6 Week 3 Speaker and Conversation Analysis

The server-side `psycon_speaker_analysis` stage uses the locally hosted `pyannote/speaker-diarization-community-1` pipeline. Regular diarization supplies overlap evidence, while exclusive diarization supplies one speaker label for transcript alignment. Faster Whisper word timestamps are assigned to the exclusive turn with at least 50% temporal overlap; unresolved boundary words remain unknown. Pyannote model access is gated and requires an operator-provisioned Hugging Face token that must never enter source control.

The system manages one consenting wearer profile. Enrollment requires exactly three clean recordings of 5–10 seconds. A SpeechBrain ECAPA-TDNN embedding is calculated for each recording, normalized, averaged, encrypted with a deployment secret, and stored without the source audio. The interface provides explicit enrollment consent, replacement, and deletion. Raw embeddings are excluded from responses and logs.

Wearer matching is verification with abstention rather than unrestricted identity search. Each diarized cluster must contain at least three seconds of clean non-overlapping speech. Exactly one cluster is labelled `participant` only when its cosine score exceeds the configured verification threshold and its advantage over the runner-up exceeds the configured ambiguity margin. Otherwise the result is `not_enrolled`, `not_identified`, `ambiguous_match`, or `insufficient_speech`; all unmatched clusters retain recording-local anonymous labels.

Per-speaker outputs include speaking duration and share, turn count and duration, articulation rate from assigned words per active-speech minute, and session speaking rate from assigned words per recording minute. Conversation timing distinguishes within-speaker pauses of at least 200 ms, positive response gaps, signed turn-transition latency, simultaneous-speech overlap, and interruptions in which a new speaker starts at least 200 ms before the active speaker ends and continues for at least 500 ms.

Standard Praat measurements include local absolute jitter, local relative jitter, RAP, PPQ5, and DDP. Measurements are made separately on continuous regions containing at least one second of quality-accepted, voiced, non-overlapping speech and are aggregated by voiced duration. Clipped, low-energy, overlapped, short, or pitch-insufficient regions return an explicit unavailable reason. Jitter is sensitive to microphone placement and noise and must not be interpreted diagnostically.

Deterministic tests use fake diarization and embedding adapters so routine CI does not download gated models. Release validation additionally requires consented multilingual multi-speaker recordings from the intended microphone, diarization error measurement, wearer false-match and false-rejection analysis, threshold calibration, and server GPU latency/memory evidence.

The local webpage accepts permission-confirmed WAV, MP3, and OGG recordings, performs in-memory decoding and balanced Protocol v2 quality windows of at most two seconds, optionally runs local transcription and speaker analysis, and displays acoustic decisions, speaker-coloured turns, attributed words, language features, conversation timing, per-speaker rates, jitter, and provenance. Separate consent is required for biometric profile comparison. Its permission checkboxes are engineering safeguards rather than a complete participant-consent system, and the local-host development server is not a production data-collection service.

7.7 Data Fusion

The backend aligns sensor and speech data using timestamps. Figure 19 shows the fusion process from sensor and speech features to a combined feature vector and the inference engine.

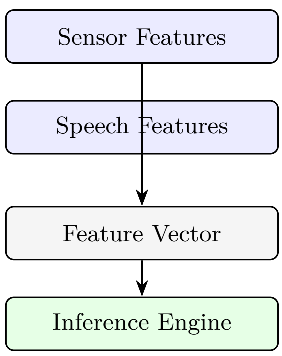

Figure 19: Data fusion: timestamp-aligned sensor and speech features are combined into a single feature vector for the inference engine.

7.8 Backend Components

Figure 20 shows the end-to-end backend pipeline, from device data ingestion to the user-facing dashboard.

7.9 Database Structure

Table 31 lists the fields stored for each session.

Table 31: Fields stored per session.

Field
User ID
Session ID
Timestamp
Sensor data
Audio metadata
Extracted features
Model output
Confidence score
Logs

Warning

For privacy, avoid storing raw audio longer than necessary unless explicit user consent and a clear research need are established. Server-side transcription requires encrypted transport and storage, authenticated access, audit logging, a documented retention and deletion schedule, and separation of participant identity from recordings and transcripts.

7.10 Machine Learning Workflow

Figure 21 shows the end-to-end machine learning workflow, from raw data collection through to inference.

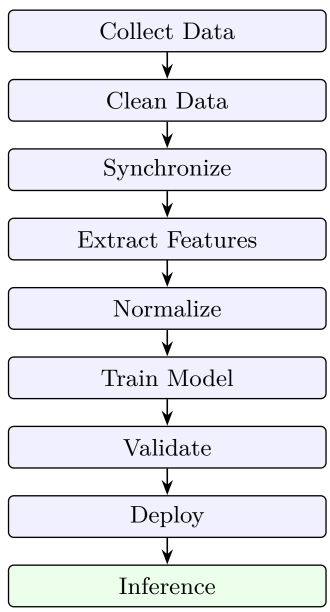

Figure 21: End-to-end machine learning workflow, from data collection and cleaning through synchronization, feature extraction, normalization, training, validation, and deployment, to inference.

7.11 Model Development

Figure 22 shows the typical model-development workflow.

Figure 20: End-to-end backend pipeline, from ESP32 device data through the API and database, feature extraction, ML model inference, and results, to the user-facing dashboard.

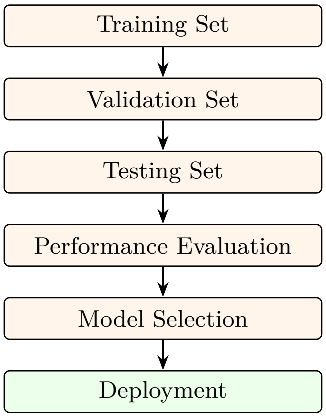

Figure 22: Typical model-development workflow, from training, validation, and testing splits through performance evaluation and model selection to deployment.

Engineering Note

Participant-level separation should be maintained between training and testing sets to reduce data leakage.

7.12 Performance Metrics

Models should be evaluated using metrics appropriate to the task, such as: accuracy; precision; recall; F1 score; ROC-AUC (for classification); and the confusion matrix.

Requirement

Confidence intervals should be reported where possible.

7.13 Prediction Output

The system should output: a predicted class or score; a confidence estimate; and data-quality indicators.

Warning

Predictions must not be presented as medical diagnoses.

7.14 Safety & Clinical Scope

Warning

PSYCON should be described as: a research prototype; a screening/research support tool; **not** a diagnostic device; and **not** a replacement for licensed mental health professionals. This distinction is important for research competitions and ethical review.

7.15 Privacy

Recommended practices: encrypt transmitted data; use anonymized participant IDs; restrict access to research data; and obtain informed consent before collecting physiological or audio data.

7.16 Model Updating

Figure 23 shows the recommended model-update lifecycle.

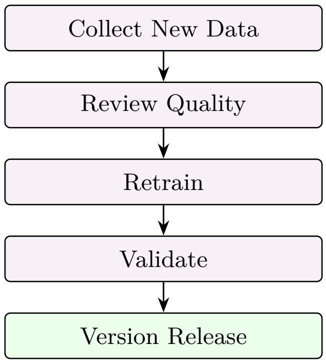

Figure 23: Recommended model-update lifecycle: collect new data, review quality, retrain, validate, and release a new version.

Design Note

Version numbers should be maintained for both datasets and models.

7.17 AI Validation Checklist

Requirement

Before deployment, the following must be verified:

- Sensor synchronization verified.
- Audio synchronization verified.
- Missing-data handling implemented.
- Feature extraction validated.
- Model evaluated on unseen data.
- Confidence outputs verified.
- False-positive/false-negative analysis completed.
- Documentation completed.

7.18 Risks

Observation

Potential technical risks:

- Motion artifacts affecting physiological signals.
- Environmental noise degrading speech quality.
- Missing packets.
- Sensor contact loss.
- Battery-related interruptions.
- Dataset imbalance.
- Model overfitting.
- Domain shift between training and real-world users.

Future Work

How each risk is mitigated should be documented as part of the AI system's risk register.

7.19 Deliverables

The final AI package should include: a data collection protocol; a dataset description; feature documentation; the model architecture; the training procedure; the validation methodology; performance metrics; version history; ethical considerations; and user documentation.

7.20 Chapter Status

Validation

Completed for this chapter:

- AI architecture
- Data pipeline
- Feature engineering overview
- Backend workflow
- Database design
- Model development process
- Validation strategy
- Privacy and ethics
- Risk assessment
- Deliverables

8 System Integration, Verification, Validation, and Deployment

8.1 System Integration Strategy

Integration occurs in five stages, illustrated in Figure 24. Every stage must pass before proceeding to the next.

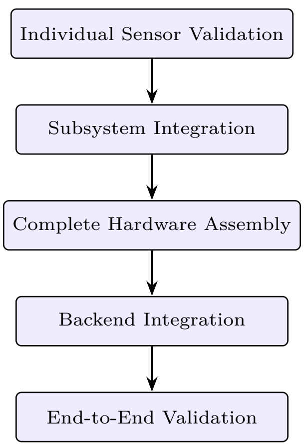

Figure 24: Five-stage PSYCON integration strategy, from individual sensor validation through to end-to-end validation.

8.2 Stage 1 — Individual Sensor Validation

Each module is tested independently before integration. Table 32 summarizes the verification checks and acceptance criteria for each sensor.

8.3 Stage 2 — I^2C Bus Validation

An I^2C scanner is run across the Wrist Module bus. The expected devices and addresses match those previously listed in Table 7.

☐ All devices detected.
☐ No duplicate addresses.
☐ No intermittent failures.

8.4 Stage 3 — Integrated Wrist Module

All wrist sensors are run simultaneously while monitoring bus stability, sampling consistency, CPU utilization, memory usage, and timing accuracy.

☐ Continuous operation without crashes.
☐ Stable timestamps.
☐ No sensor starvation.

8.5 Stage 4 — Integrated Audio Module

Verification covers continuous microphone capture, ambient light acquisition, and concurrent wireless communication.

☐ No dropped audio buffers.
☐ Stable light sensor readings.

8.6 Stage 5 — Full System Integration

Both modules operate together, with verification of time synchronization, data transmission, backend reception, and session management.

☐ Complete synchronized datasets.
☐ No packet corruption.

8.7 Functional Testing

Functional testing confirms that the system can:

☐ Boot successfully.
☐ Detect all sensors.
☐ Acquire data.
☐ Timestamp correctly.
☐ Transmit data.
☐ Recover from temporary communication loss.
☐ Shut down safely.

8.8 Stress Testing

Tests conducted.

- 1-hour continuous operation.
- Extended operation (target duration based on battery capacity).
- Rapid reboot cycles.
- Repeated connection/disconnection tests.
- Continuous wireless transmission.

Monitored quantities.

- Temperature.
- Memory usage.
- Packet loss.
- Unexpected resets.

8.9 Power Validation

Verification.

- Battery charging.
- Battery discharge.

Table 32: Stage 1 individual sensor validation: verification checks and acceptance criteria.

Sensor	Verification Checks	Acceptance Criteria
MAX30102	I^2C communication; raw RED and IR readings; finger detection; stable waveform; no communication errors	Device detected at `0x57`; stable signal; no bus failures
MPU6050	Accelerometer; gyroscope; temperature register; axis orientation	Address `0x68`; stable readings at rest; motion reflected correctly
MCP9808	Ambient temperature; stable readings; repeatability	Address `0x18`; no sudden fluctuations
ADS1115	Address detection; all four channels readable; stable ADC values	Address `0x48`; no random saturation
TEMT6000	Analog response to light; dark vs. bright environments	Smooth output variation
INMP441	Audio recording; no clipping; correct I^2S configuration; acceptable noise floor	Clean recordings; no DMA overruns

- Low-battery detection.
- Brownout handling.
- Controlled shutdown.

Recorded quantities.

- Battery voltage.
- Runtime.
- Charging time.
- Device temperature.

8.10 Sensor Calibration

Calibration procedures are documented for MAX30102, GSR, MCP9808, TEMT6000, and MPU6050. Each calibration record should include:

- Calibration date.
- Firmware version.
- Calibration method.
- Environmental conditions.

8.11 Software Validation

☐ Firmware boots correctly.
☐ Sensor drivers initialize.
☐ Communication stack functions.
☐ Logging system operates.
☐ Error recovery works.

8.12 AI Validation

☐ Feature extraction.
☐ Timestamp synchronization.
☐ Inference pipeline.
☐ Model confidence reporting.
☐ Handling of missing or corrupted data.

8.13 Documentation Package

The complete documentation package should include:

☐ Bill of Materials (BOM).
☐ Wiring diagrams.
☐ GPIO map.
☐ Power tree.
☐ Firmware architecture.
☐ AI architecture.
☐ Testing reports.
☐ Calibration records.
☐ Risk assessment.
☐ User manual.
☐ Assembly guide.
☐ Version history.

8.14 Competition Readiness Checklist

Requirement

Before submission:

☐ Hardware fully assembled.
☐ Firmware stable.
☐ AI pipeline validated.
☐ Documentation complete.
☐ Safety review completed.
☐ Demonstration rehearsed.
☐ Backup firmware available.
☐ Backup hardware available (where practical).

8.15 Known Remaining Risks

Observation

Current engineering risks include:

- GSR analog front-end refinement.
- Battery runtime limitations.
- Wireless reliability in congested environments.
- Motion artifacts affecting physiological signals.
- Audio quality in noisy environments.
- Long-term sensor stability.

Each risk should have a documented mitigation strategy.

8.16 Final Deliverables

The complete PSYCON package should contain:

- Hardware prototype.
- Embedded firmware.
- AI models.
- Backend software.
- Mobile/web interface (if applicable).

- Technical documentation.
- Validation and testing reports.
- User and maintenance guides.
- Safety and ethics documentation.
- Research report.

8.17 Volume 1 Summary

Validation

This concludes the core engineering design document, covering:

- Project overview
- Hardware architecture
- Electrical design
- Firmware architecture
- AI pipeline
- System integration
- Verification and validation

Part II
Volume 2: Manufacturing, Mechanical Design & Future Roadmap

9 Manufacturing, PCB Design, and Mechanical Engineering

9.1 Manufacturing Philosophy

PSYCON is designed as a research prototype with a clear migration path, shown in Figure 25. The current implementation emphasizes rapid prototyping, validation, and documentation before hardware miniaturization.

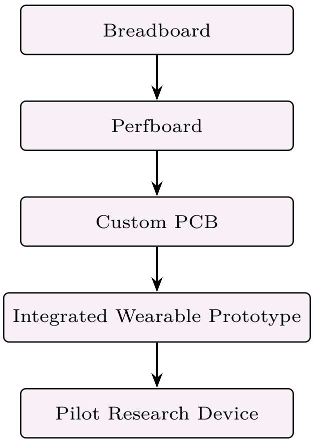

Figure 25: PSYCON manufacturing migration path, from breadboard prototyping through to a pilot research device.

9.2 PCB Design Goals

The custom PCB should:

- Reduce wiring complexity.
- Improve signal integrity.
- Lower power consumption.
- Reduce mechanical failure points.
- Simplify assembly.
- Improve repeatability across multiple units.

9.3 PCB Architecture

Table 33 summarizes which components are integrated on the Wrist PCB versus the Audio PCB.

9.4 Mechanical Layout

Figure 26 shows the suggested mechanical stack-up for both modules.

Wrist Module. The optical sensor should remain flush against the skin, while electronics are isolated from direct skin contact.

Audio Module. The microphone opening should remain unobstructed and protected from dust.

9.5 Enclosure Requirements

- Lightweight.
- Rounded edges.
- Sweat resistant.
- Battery securely retained.
- Accessible USB charging port.
- Reset/programming access.
- Ventilation where required.
- Strain relief for any external wiring.

9.6 Assembly Process

1. Inspect all PCBs.
2. Verify battery polarity.
3. Assemble power system.
4. Program ESP32 firmware.
5. Test each sensor individually.
6. Integrate all modules.
7. Perform system-level verification.
8. Close enclosure.
9. Final functional test.

9.7 Quality Control Checklist

Before final assembly:

☐ No solder bridges.
☐ Correct component orientation.
☐ Secure connectors.
☐ Proper insulation.
☐ Stable power rails.
☐ All sensors detected.
☐ Firmware version recorded.

9.8 Maintenance

Recommended periodic checks:

- Battery health.
- Connector wear.
- Sensor cleanliness.
- Enclosure integrity.
- Firmware updates.
- Calibration verification.

9.9 Reliability Considerations

Observation

Potential failure modes:

- Battery degradation.
- Loose connectors.
- Sensor contamination.
- Wire fatigue.
- Moisture ingress.
- Firmware corruption.
- Electrostatic discharge (ESD).

Documented mitigation should be provided for each failure mode.

Table 33: PCB architecture: component allocation between the Wrist PCB and Audio PCB.

Component	Wrist PCB	Audio PCB
ESP32	✓	✓
MAX30102	✓	—
ADS1115	✓	—
MCP9808	✓	—
MPU6050	✓	—
GSR analog front-end	✓	—
INMP441	—	✓
TEMT6000	—	✓
Battery connector	✓	✓
Charging interface	✓	✓
Programming header	✓	✓
Power regulation	✓	—
Status LEDs	—	✓

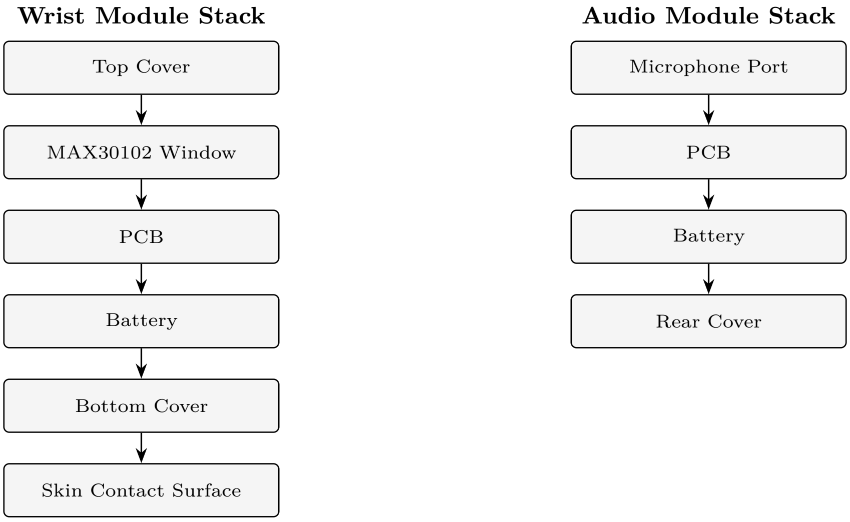

Figure 26: Suggested mechanical stack-up for the Wrist Module (top cover through skin-contact surface) and Audio Module (microphone port through rear cover).

9.10 Future Improvements

Future Work

Potential upgrades:

- Custom low-power PCB.
- Dedicated medical-grade GSR analog front-end.
- Lower-power MCU variants.
- Larger-capacity battery with improved power management.
- Secure data encryption hardware.
- OTA firmware updates.
- Additional physiological sensors (only after validating clinical usefulness).
- Improved enclosure with waterproofing.

9.11 Scalability

Future versions may support:

- Multiple wearable form factors.
- Cloud synchronization.
- Long-duration monitoring.
- Clinical research dashboards.
- Multi-user study management.
- Automated firmware update system.

9.12 Final Manufacturing Deliverables

- ☐ Complete Bill of Materials (BOM).
- ☐ PCB design files.
- ☐ Schematic diagrams.
- ☐ Assembly drawings.
- ☐ Wiring diagrams.
- ☐ Mechanical CAD files.
- ☐ Enclosure STL files.
- ☐ Firmware source code.
- ☐ AI model documentation.
- ☐ Test reports.
- ☐ Validation records.
- ☐ User documentation.
- ☐ Risk assessment.

9.13 Volume 2 Status

Validation

Completed for this chapter:

- Manufacturing strategy
- PCB migration plan
- Mechanical architecture
- Enclosure requirements
- Assembly workflow
- Quality control
- Maintenance plan
- Reliability considerations
- Future roadmap

Part III

Volume 3: Reference Documentation

10 Complete Bill of Materials

10.1 Wrist Module BOM

Table 34 lists the complete bill of materials for the Wrist Module.

10.2 Audio Module BOM

Table 35 lists the complete bill of materials for the Audio Module.

10.3 Shared Tools

- Breadboard
- Soldering iron
- Solder wire
- Flux
- Heat-shrink tubing
- USB-C cable
- ESP32 programming cable
- Small screwdrivers
- Tweezers
- Hot glue/Kapton tape for strain relief

10.4 Electrical Specifications

Logic voltage. ESP32 GPIO: 3.3 V. All digital interfaces: 3.3 V.

Battery. Nominal: 3.7 V. Fully charged: 4.2 V. Recommended cutoff: ≈3.2–3.3 V under light load.

Communication. I^2C; I^2S; Wi-Fi; BLE.

11 Complete GPIO Assignment

11.1 Wrist Module

Table 36 lists the complete GPIO assignment for the Wrist Module, and Table 37 lists the corresponding I^2C device addresses.

Table 36: Wrist Module GPIO assignment (reference summary).

ESP32 Pin	Device
GPIO21	I^2C SDA
GPIO22	I^2C SCL
GPIO34	GSR (ADS1115 input interface)
GPIO35	Battery monitoring (if used)
3V3	Sensors
GND	Common ground

Table 37: I^2C device addresses (reference summary).

Device	Address
MCP9808	`0x18`
ADS1115	`0x48`
MAX30102	`0x57`
MPU6050	`0x68`

11.2 Audio Module

Table 38 lists the complete GPIO assignment for the Audio Module.

Table 38: Audio Module GPIO assignment (reference summary).

ESP32 Pin	Device
GPIO21	I^2C SDA (if required)
GPIO22	I^2C SCL (if required)
GPIO25	I^2S WS
GPIO26	I^2S SCK
GPIO33	I^2S SD
GPIO32	TEMT6000 analog input
3V3	Sensor power
GND	Ground

12 Wiring Rules

12.1 Always

- Share one common ground.
- Keep I^2C wires short.
- Keep microphone traces away from power lines.
- Separate analog GSR wiring from digital signals.
- Twist long GSR leads if possible.
- Secure battery wiring to prevent movement.

12.2 Avoid

Warning

- 5 V directly on ESP32 GPIOs.
- Charging while electrodes are attached.
- Long unshielded microphone wires.
- Loose JST connectors.

13 Calibration Summary

For each sensor, the following should be documented:

- Calibration date
- Firmware version
- Environmental conditions
- Test operator
- Calibration method
- Results

Requirement

Recalibration is required after hardware modifications or sensor replacement.

Table 34: Wrist Module bill of materials.

Qty.	Component	Purpose
1	ESP32 DevKit V1 (30-pin)	Main controller
1	MAX30102	Heart rate & PPG
1	ADS1115	16-bit ADC for GSR
1	MPU6050	Motion sensing
1	MCP9808	Temperature sensing
1 pair	GSR electrodes	Skin conductance measurement
1	TP4056 USB-C charger (protected)	Battery charging
2	500 mAh LiPo cells (parallel)	Power source
1	Power switch	Main power
1	JST connector	Battery connection
Several	Male/female headers	Module connections
Several	Hook-up wires	Wiring
1	Wrist enclosure	Housing

Table 35: Audio Module bill of materials.

Qty.	Component	Purpose
1	ESP32 DevKit V1 (30-pin)	Main controller
1	INMP441	I^2S microphone
1	TEMT6000	Ambient light sensor
1	TP4056 USB-C charger (protected)	Battery charging
1	500 mAh LiPo cell	Power source
1	Power switch	Main power
1	JST connector	Battery connection
Several	Hook-up wires	Wiring
1	Enclosure	Housing

14 Test Log Template

Table 39 lists the fields to be recorded for every build.

Table 39: Test log template fields.

Field
Hardware revision
Firmware version
Date
Tester
Sensor initialization results
I^2C scan results
Battery voltage
Runtime
Charging time
Communication status
Notes and corrective actions

15 Risk Register

Table 40 consolidates the project's known risks, their impact level, and the corresponding mitigation strategy.

16 Project Deliverables

The final PSYCON package should include:

- ☐ Hardware prototype
- ☐ Firmware source code
- ☐ Wiring diagrams
- ☐ Bill of Materials (BOM)
- ☐ Schematics
- ☐ AI documentation
- ☐ Testing reports
- ☐ Validation reports
- ☐ Risk assessment
- ☐ User manual
- ☐ Assembly guide
- ☐ Maintenance guide
- ☐ Version history
- ☐ Competition report
- ☐ Demonstration material

16.1 Volume 3 Status

Validation

Completed for this volume:

- Complete BOM
- GPIO assignments
- Electrical specifications
- Wiring rules

Table 40: PSYCON risk register.

Risk	Impact	Mitigation
Battery depletion	HIGH	Low-battery detection & controlled shutdown
Sensor disconnect	MEDIUM	Startup/self-test & continuous monitoring
I^2C communication failure	MEDIUM	Retry and error logging
Motion artifacts	HIGH	Filtering and quality metrics
Audio noise	MEDIUM	Mechanical isolation and preprocessing
GSR electrode contact loss	HIGH	Contact-quality checks and proper placement
Firmware crash	HIGH	Watchdog timer and restart
Wireless interruption	MEDIUM	Packet buffering and retry
Data corruption	HIGH	Checksums and sequence numbers

- Calibration documentation
- Test logging template
- Risk register
- Final deliverables

Part IV
Volume 4: Research Methodology, Ethics & Competition Documentation

17 Research Methodology

17.1 Research Objective

The objective of PSYCON is to investigate whether multimodal physiological and speech-derived features can be combined to estimate indicators relevant to psychological well-being in a research setting. The system is intended as a research and screening prototype, not a diagnostic medical device.

17.2 Research Questions

Primary questions:

- Can physiological and speech features improve predictive performance compared with either modality alone?
- How robust are predictions under different environmental conditions?
- How does motion affect signal quality?
- What is the minimum recording duration needed for stable feature extraction?

17.3 Study Design

Figure 27 shows the suggested end-to-end study workflow.

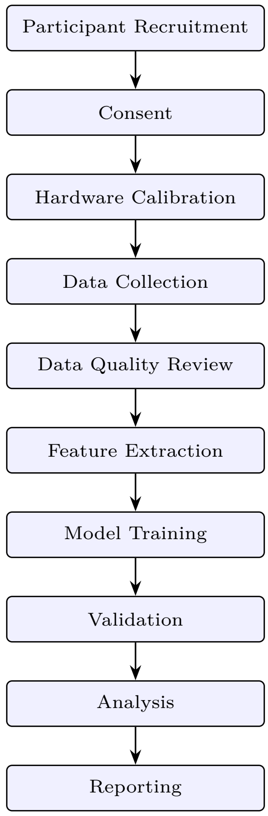

Figure 27: Suggested end-to-end study workflow, from participant recruitment and consent through data collection, feature extraction, model training and validation, to final analysis and reporting.

17.4 Participants

For each participant, the following should be recorded:

- Anonymous participant ID
- Age group (if relevant)
- Biological sex (optional, if ethically approved)
- Recording date/time
- Session duration
- Environmental conditions
- Device firmware version

Warning

Avoid collecting personally identifying information unless necessary and approved.

17.5 Data Collection Protocol

For every session:

1. Verify hardware operation.
2. Confirm battery level.
3. Attach wrist module.
4. Position audio module.
5. Start synchronized recording.
6. Monitor system status.
7. End recording.
8. Save data.
9. Backup data.
10. Review data quality.

17.6 Recorded Variables

Table 41 lists the variables recorded for each session, organized by category.

Table 41: Recorded variables by category.

Category	Variables
Physiological	Heart rate; PPG waveform; GSR; temperature; motion; ambient light; battery status
Speech	Audio recording (if consented); extracted acoustic features; extracted language features
System	Timestamp; session ID; firmware version; sensor status; error logs

17.7 Inclusion Criteria

Example criteria:

- Adult participants (or appropriate guardian consent for minors).
- Ability to provide informed consent.
- Ability to wear the device comfortably.

17.8 Exclusion Criteria

Example criteria:

- Damaged skin at electrode location.
- Implanted medical devices, if required by the safety review.
- Participants unwilling to consent to physiological or audio recording.

18 Data Management

18.1 Storage

The following should be maintained:

- Raw sensor files
- Processed feature files
- Metadata
- Model outputs
- Logs

Engineering Note

Use regular backups and version control.

18.2 File Structure

Listing 2 shows the recommended top-level data repository layout.

```
/participants
/sessions
  /raw
  /features
/models
/results
/docs
/firmware
```

Listing 2: Recommended PSYCON data repository structure.

18.3 Version Control

The following should be tracked:

- Firmware version
- Hardware revision
- PCB revision
- Dataset version
- Model version
- Documentation revision

19 Ethics and Privacy

19.1 Ethical Principles

- Voluntary participation.
- Informed consent.
- Right to withdraw.
- Data minimization.
- Secure storage.
- Limited access.
- Transparent reporting.

Requirement

If recording speech, clearly explain what is collected, how it is processed, how long it is retained, and who can access it.

19.2 Wearer Voice Profile

Speaker verification is biometric processing and requires consent separate from permission to process the conversation recording. Week 3 supports one local wearer profile created from three quality-checked recordings. Only the normalized averaged voice embedding and non-identifying enrollment metadata may be retained; the source enrollment recordings must be discarded immediately after processing. The profile must be encrypted using an operator-provisioned secret, excluded from logs and API responses, and replaceable or deletable by the operator.

The profile may identify only the enrolled wearer. Other voices remain anonymous recording-local speaker labels, and an uncertain comparison must abstain rather than select the nearest voice. Production deployment additionally requires authenticated access, encrypted transport, audit logging, a documented retention period, and measured false-match and false-rejection rates across the intended languages, microphones, and environments.

20 Statistical Analysis Plan

Potential analyses include:

- Descriptive statistics
- Correlation analysis
- Classification metrics
- Cross-validation
- Ablation studies (speech only vs. physiology only vs. combined)
- Error analysis
- Confusion matrices
- ROC curves (where applicable)

Observation

Both overall performance and limitations should be reported.

21 Competition Documentation

The following should be prepared:

- ☐ Executive summary
- ☐ Abstract
- ☐ Problem statement
- ☐ Innovation overview
- ☐ Hardware architecture
- ☐ Software architecture
- ☐ AI methodology
- ☐ Testing results
- ☐ Limitations
- ☐ Future work
- ☐ References
- ☐ Safety statement
- ☐ Ethics statement
- ☐ Budget
- ☐ Timeline

22 Demonstration Plan

A live demonstration should be prepared, showing:

- Device startup.
- Sensor initialization.
- Live physiological acquisition.
- Live audio capture (with consent).
- Data synchronization.
- Backend visualization.
- AI inference output.
- Consented wearer enrollment and encrypted-profile deletion.
- Anonymous speaker timeline, conservative wearer match, and attributed transcript.
- Per-speaker speaking rates, pauses, response gaps, overlaps, interruptions, and jitter with unavailable reasons.
- Safe shutdown.

Requirement

A recorded backup demonstration should also be prepared in case of hardware issues.

The Week 3 software demonstration uses a consented multilingual recording with at least two speakers, natural pauses, and overlap. It must show that missing model access, missing enrollment, ambiguous identity, and poor-quality speech produce explicit abstention states while acoustic analysis and transcription remain independently usable.

23 Project Limitations

Current limitations include:

- Prototype hardware rather than custom PCB.
- Battery life constrained by ESP32 development boards.
- GSR accuracy dependent on analog front-end implementation.
- Performance depends on the quality and diversity of training data.

Warning

Outputs are intended for research support and must not be interpreted as clinical diagnoses.

23.1 Volume 4 Status

Validation

Completed for this volume:

- Research methodology
- Study protocol
- Data management
- Ethics and privacy
- Statistical analysis plan
- Competition documentation
- Demonstration plan
- Limitations

Part V
Volume 5: Appendices, Reference Material & Project Closure

24 Datasheet and Reference Index

The references in Table 42 should be maintained as part of the project documentation. Official manufacturer datasheets should be used where available.

25 Communication Protocol

25.1 Wrist ESP32 to Backend and Audio ESP32 to Backend

Figure 28 shows the field structure of each transmitted packet for the Wrist and Audio modules.

25.2 Synchronization

Both ESP32 modules should maintain synchronized timestamps. Figure 29 shows the recommended synchronization sequence.

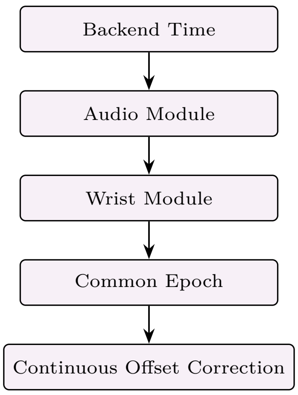

Figure 29: Recommended synchronization sequence, from backend time distribution through per-module alignment to a common epoch with continuous offset correction.

26 Firmware Architecture

Figure 30 shows the boot-to-shutdown sequence for the Wrist and Audio firmware.

26.1 Wrist Firmware Tasks

- Power management
- Sensor polling
- Timestamp generation
- Error logging
- Battery monitoring
- Packet transmission
- Watchdog servicing

27 Backend Software Architecture

Figure 31 shows the end-to-end backend software pipeline. Modules should be independently testable and version-controlled.

28 Complete Test Procedures

Table 43 consolidates the complete set of test procedures across hardware, electrical, firmware, AI, and mechanical domains.

29 Maintenance Schedule

Table 44 lists recommended maintenance tasks by frequency.

Table 44: Maintenance schedule by frequency.

Frequency	Tasks
Before every use	Check battery charge; inspect wiring; clean optical sensor; clean GSR electrodes; verify microphone opening
Weekly	Inspect solder joints; verify charging; check firmware version; review logs
Monthly	Recalibrate sensors (if applicable); battery health review; enclosure inspection; connector inspection

30 Troubleshooting Guide

Table 45 lists common symptoms, together with the possible causes or checks to perform.

31 Glossary

Table 46 defines the abbreviations and terms used throughout this document.

32 Revision History

Table 47 lists the document revision history.

Table 42: Datasheet and reference index.

Component	Primary Documentation
ESP32-WROOM-32	Espressif ESP32 datasheet & technical reference manual
MAX30102	Maxim Integrated datasheet
MPU6050	InvenSense register map & datasheet
ADS1115	Texas Instruments ADS1115 datasheet
MCP9808	Microchip MCP9808 datasheet
INMP441	InvenSense INMP441 datasheet
TEMT6000	Vishay TEMT6000 datasheet
TP4056	TP4056 charger IC datasheet
DW01A	Battery protection IC datasheet
FS8205A	Dual MOSFET datasheet

Table 43: Complete test procedures by category.

Category	Test Items
Hardware	Visual inspection; connector inspection; battery polarity verification; sensor detection; charging verification; thermal inspection
Electrical	Supply voltage validation; brownout verification; current consumption; runtime testing; charging cycle testing
Firmware	Boot reliability; sensor initialization; error recovery; watchdog operation; memory stability; long-duration operation
AI	Dataset integrity; feature extraction validation; training reproducibility; cross-validation; performance evaluation; error analysis
Mechanical	Wearability; strap integrity; enclosure fit; cable strain relief; button accessibility; charging access

Table 47: Document revision history.

Version	Description
v0.1	Initial concept
v0.2	Hardware selection
v0.3	Sensor integration plan
v0.4	Firmware architecture
v0.5	AI pipeline definition
v0.6	Validation strategy
v0.7	Mechanical design
v0.8	Competition documentation
v0.9	Prototype integration
v1.0	First complete engineering release
v1.1	Added the implemented Week 3 audio feature, provenance, quality-decision, fixture, and demo baseline
v1.2	Added consent-gated local transcription, English language/conversation features, and the real-recording webpage
v1.3	Selected small.en for local development and large-v3-turbo for GPU-backed server transcription
v1.4	Switched local transcription to multilingual small and clarified non-English feature abstention
v1.5	Completed the versioned pitch-statistics, pause-duration, and voice-stability feature contract

33 Final Project Checklist

33.1 Hardware

- ✓ Component selection finalized
- ✓ Power architecture defined
- ✓ Sensor interfaces defined
- ✓ GPIO allocation documented
- ✓ I^2C address map documented
- ☐ Final integrated assembly
- ☐ Electrical validation
- ☐ Runtime testing

33.2 Firmware

- ✓ Architecture defined
- ✓ Communication protocol defined
- ☐ Complete implementation
- ☐ End-to-end testing
- ☐ Performance optimization

33.3 AI

- ✓ Pipeline defined
- ✓ Feature plan documented
- ☐ Dataset collection
- ☐ Model training
- ☐ External validation

33.4 Documentation

- ✓ System architecture
- ✓ Bill of Materials (BOM)

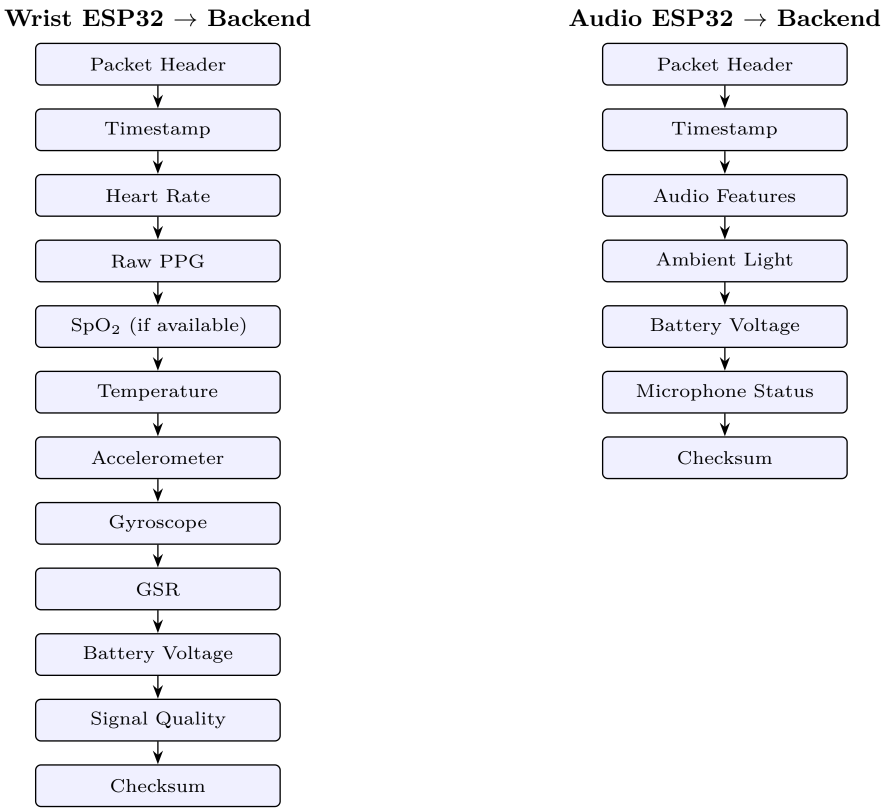

Figure 28: Field structure of packets transmitted from the Wrist ESP32 (left, 12 fields) and Audio ESP32 (right, 7 fields) to the backend.

Table 45: Troubleshooting guide: symptoms and possible causes or checks.

Symptom	Possible Causes / Checks
Device does not boot	Battery discharged; incorrect polarity; loose connector; faulty regulator; firmware corruption
Sensor not detected	SDA; SCL; pull-ups; address conflicts; power supply; ground continuity
No audio	I^2S wiring; pin assignments; gain settings; DMA configuration; buffer overflow
Poor GSR signal	Electrode contact; motion artifacts; wiring; ADC configuration; analog front end
Random ESP32 reset	Brownout; insufficient power; memory exhaustion; watchdog timeout; software crash

- ✓ Wiring plan
- ✓ Risk assessment
- ✓ Research methodology
- ✓ Ethics documentation
- ✓ Maintenance plan
- ✓ Troubleshooting guide
- ✓ Reference index
- ✓ Competition documentation

34 Conclusion

PSYCON is a modular research platform designed to investigate whether physiological measurements and speech-derived features can be combined to estimate indicators relevant to psychological well-being. Its split wearable architecture, documented hardware stack, and planned AI pipeline provide a structured foundation for research and engineering development.

To reach a competition-ready prototype, the remaining priorities are completing firmware integration, validating electrical performance and battery behavior, refining the GSR analog front end, collecting a representative dataset, and rigorously evaluating model performance.

Warning

Throughout development and presentation, the system should be described as a research and screening prototype, not as a medical diagnostic device.

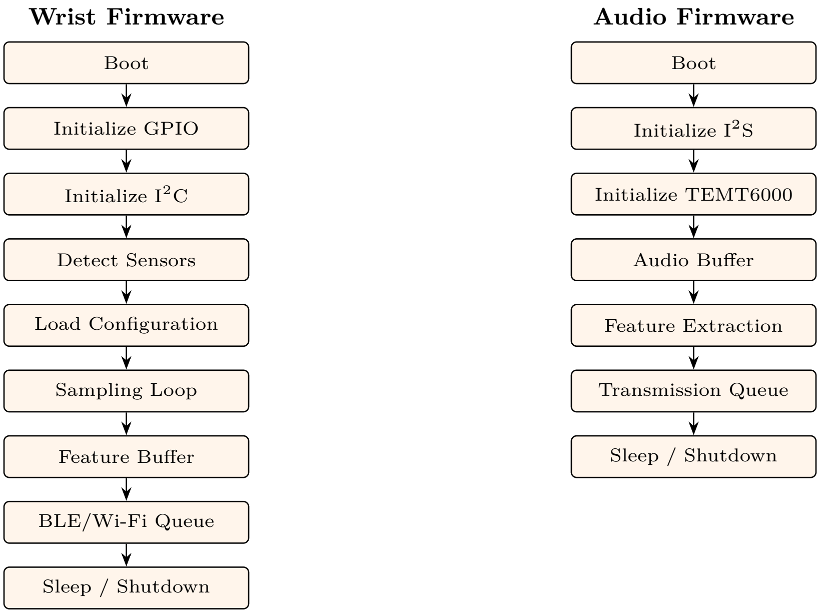

Figure 30: Boot-to-shutdown sequence for the Wrist firmware (left, 9 stages) and Audio firmware (right, 7 stages).

Figure 31: End-to-end backend software pipeline, from API gateway and authentication through packet validation, time synchronization, storage, feature extraction, AI inference, visualization, and research export.

Table 46: Glossary of terms and abbreviations.

Term	Definition
ADC	Analog-to-Digital Converter
BLE	Bluetooth Low Energy
DMA	Direct Memory Access
EDA/GSR	Electrodermal Activity / Galvanic Skin Response
ESP32	Dual-core Wi-Fi/Bluetooth microcontroller
HRV	Heart Rate Variability
I^{2}C	Inter-Integrated Circuit communication bus
I^{2}S	Integrated Interchip Sound interface
IMU	Inertial Measurement Unit
LiPo	Lithium Polymer battery
MCU	Microcontroller Unit
MFCC	Mel-Frequency Cepstral Coefficients
PCB	Printed Circuit Board
PPG	Photoplethysmography
SpO_2	Peripheral oxygen saturation
UART	Universal Asynchronous Receiver Transmitter